Training the CNN

Testing sages

- epochs: train longer than 10 and see if performance improves
- learning rate: if training is unstable or stalls, adjust the optimizer learning rate
- batch size: try something like 16, 32, 64
- dropout rate: test values like 0.2, 0.3, 0.5
- decision threshold: don't assume 0.5 is best, especially for medical classification

That order is useful because these are easier and faster to test than redesigning the CNN.

Best first sequence:

1. more epochs
2. threshold tuning
3. batch size
4. dropout
5. learning rate

Model Architecture 1:

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 64)	0
dense (Dense)	(None, 128)	8,320
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 1)	129

10 Epochs

Training

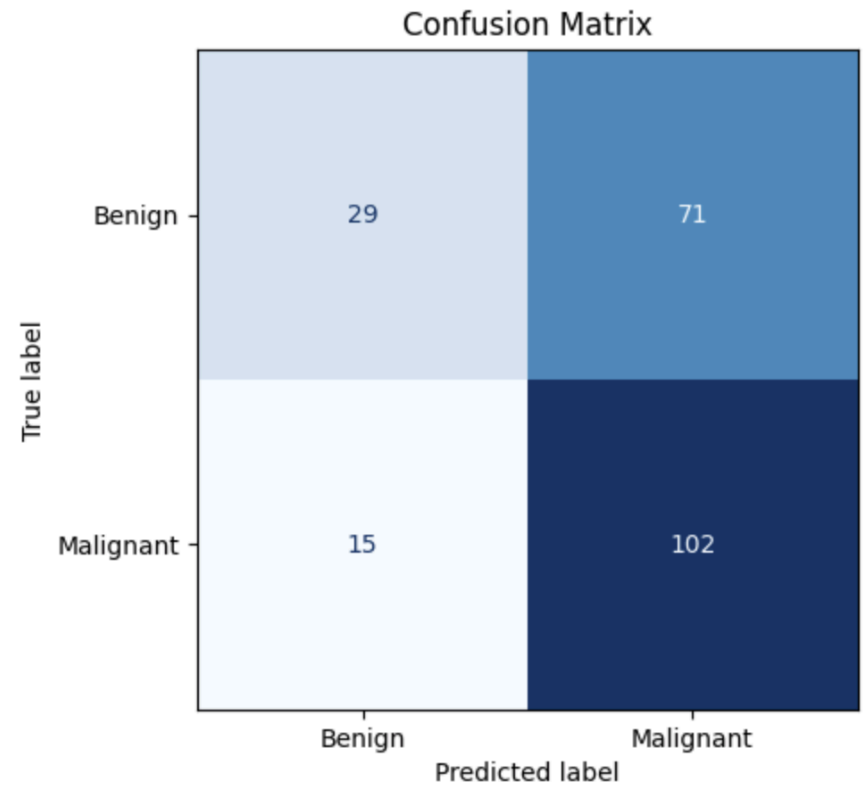
```
Epoch 1/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 29s 712ms/step - accuracy: 0.4971 - auc: 0.4812 - loss: 3.8407 - precision: 0.5342 - recall: 0.5182
Epoch 2/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 5s 176ms/step - accuracy: 0.5502 - auc: 0.5472 - loss: 1.0235 - precision: 0.5842 - recall: 0.5717
Epoch 3/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 4s 140ms/step - accuracy: 0.5894 - auc: 0.5847 - loss: 0.6961 - precision: 0.6018 - recall: 0.7024
Epoch 4/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 4s 139ms/step - accuracy: 0.5859 - auc: 0.6028 - loss: 0.6781 - precision: 0.6055 - recall: 0.6638
Epoch 5/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 4s 150ms/step - accuracy: 0.6217 - auc: 0.6488 - loss: 0.6603 - precision: 0.6355 - recall: 0.6981
Epoch 6/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 4s 140ms/step - accuracy: 0.6171 - auc: 0.6518 - loss: 0.6615 - precision: 0.6316 - recall: 0.6938
Epoch 7/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 4s 144ms/step - accuracy: 0.6194 - auc: 0.6510 - loss: 0.6661 - precision: 0.6315 - recall: 0.7045
Epoch 8/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 4s 147ms/step - accuracy: 0.6494 - auc: 0.6894 - loss: 0.6402 - precision: 0.6564 - recall: 0.7323
Epoch 9/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 4s 144ms/step - accuracy: 0.6321 - auc: 0.6853 - loss: 0.6389 - precision: 0.6623 - recall: 0.6467
Epoch 10/10
28/28 ━━━━━━━━━━━━━━━━━━━━ 4s 139ms/step - accuracy: 0.6355 - auc: 0.6772 - loss: 0.6450 - precision: 0.6406 - recall: 0.7366
dict_keys(['accuracy', 'auc', 'loss', 'precision', 'recall'])
```

Evaluation

Evaluation on testing set - threshold of 0.5

```
'accuracy': 0.6036866307258606,
'auc': 0.6342307925224304,
'loss': 0.6984637975692749,
'precision': 0.589595377445221,
'recall': 0.8717948794364929
```

Confusion Matrix

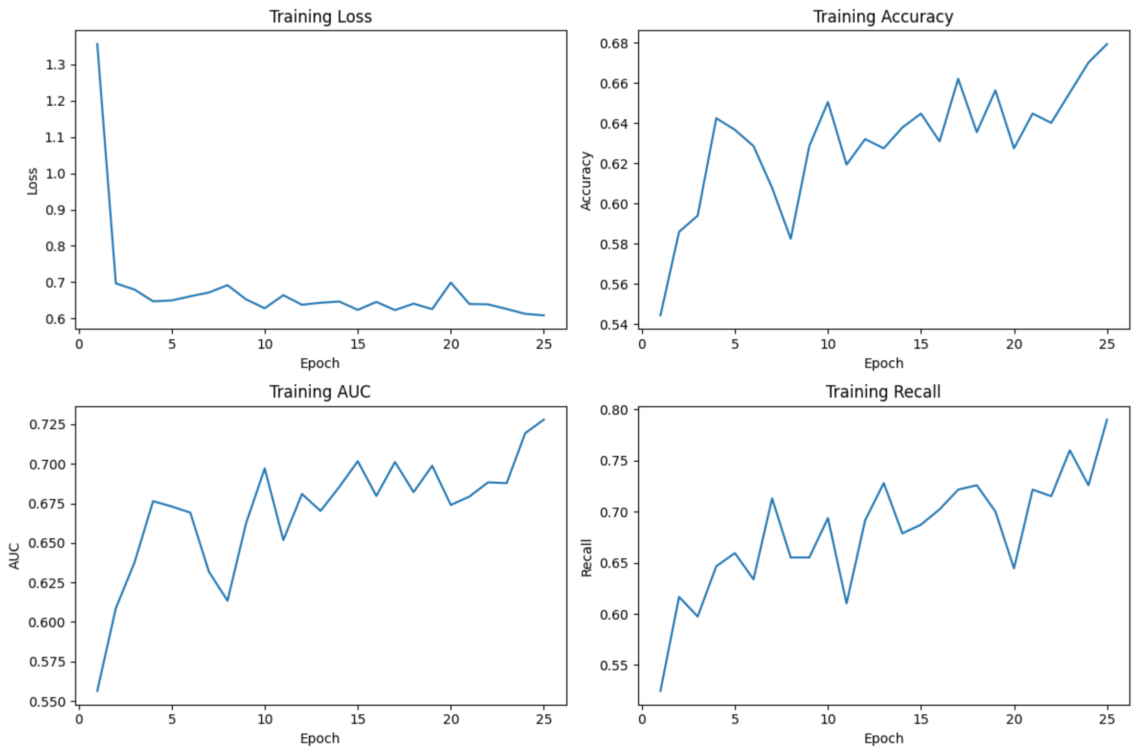


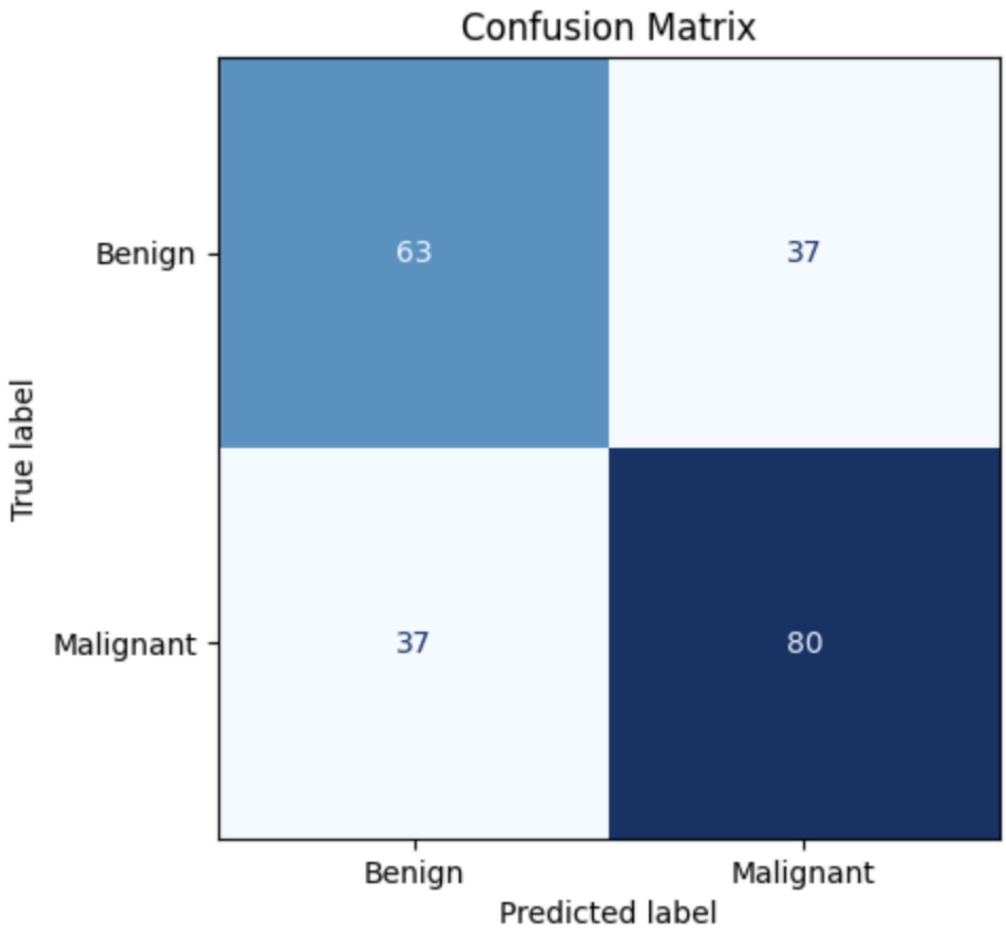
Overall

- The model catches most Malignat cases
- Good recall for malignant is pretty strong: `102 / (102 + 15) = 87.2%`
    - **Recall is often more important, missing a malignant case is costly**
- It is over predicting malignant quite a bit
    - Benign recall is low: `29 / (29 +71) = 29%`

25 Epochs

accuracy: 0.6589861512184143,
auc: 0.7073931694030762,
loss: 0.6601645350456238,
precision: 0.6837607026100159,
recall: 0.6837607026100159

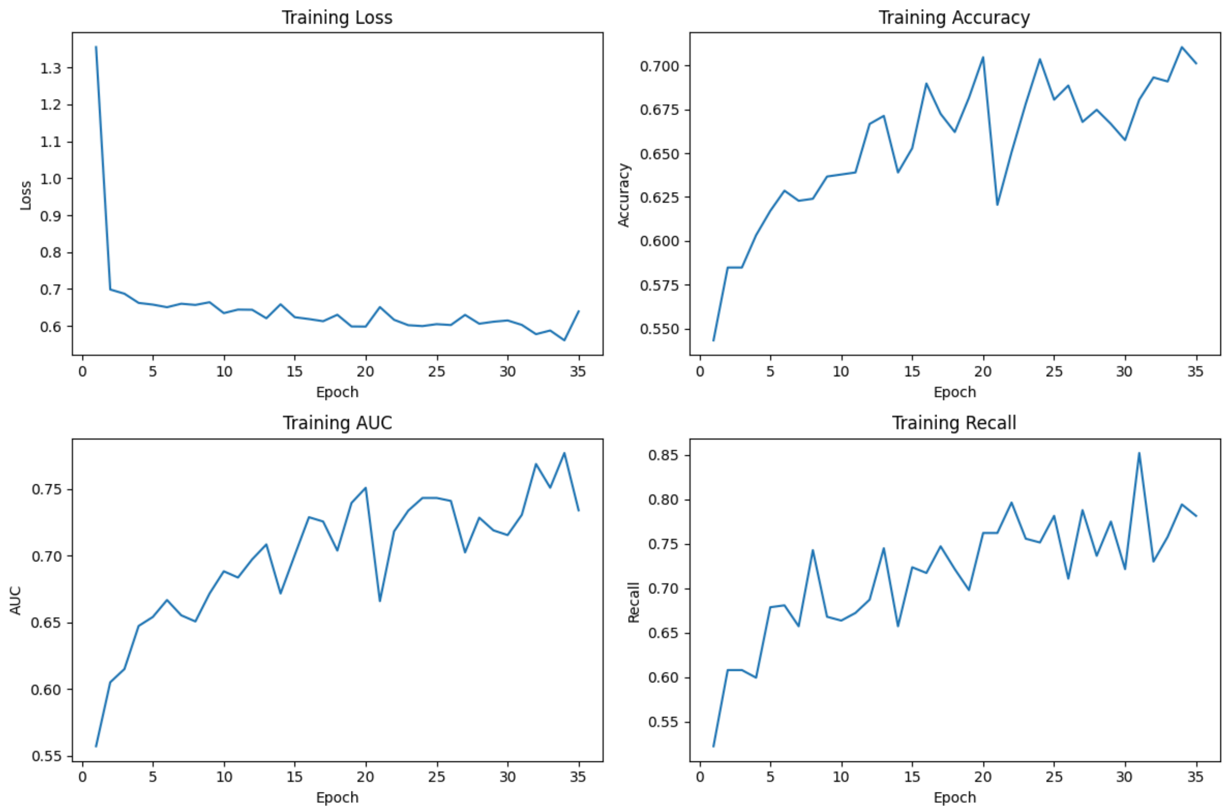

Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch

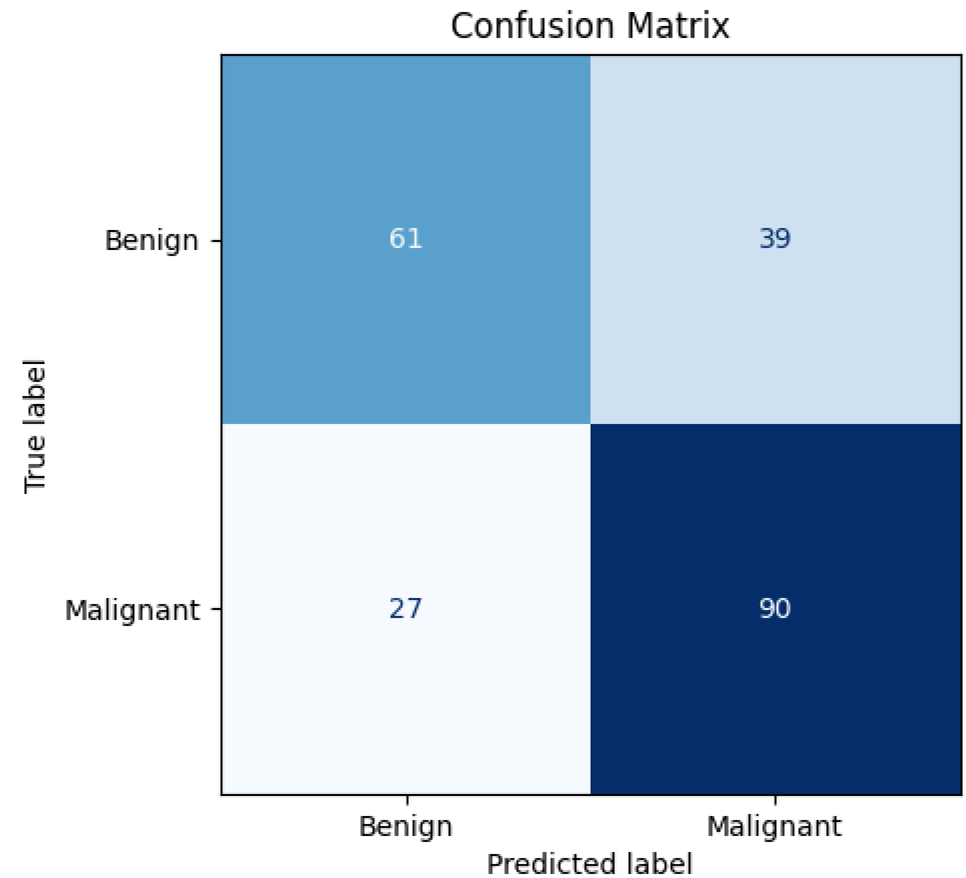


	precision	recall	f1-score	support
Benign	0.63	0.63	0.63	100
Malignant	0.68	0.68	0.68	117
accuracy			0.66	217
macro avg	0.66	0.66	0.66	217
weighted avg	0.66	0.66	0.66	217

35 Epochs

accuracy: 0.695852518081665,
auc: 0.7087180018424988,
loss: 0.7986871600151062,
precision: 0.6976743936538696,
recall: 0.7692307829856873

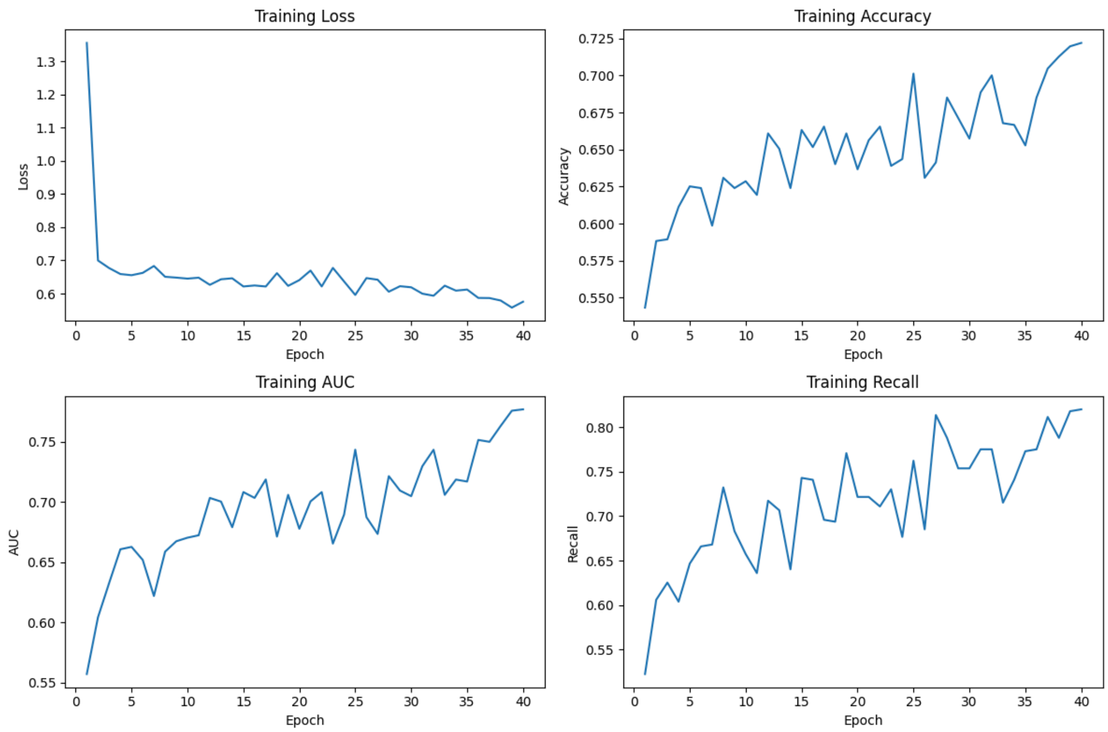

Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch

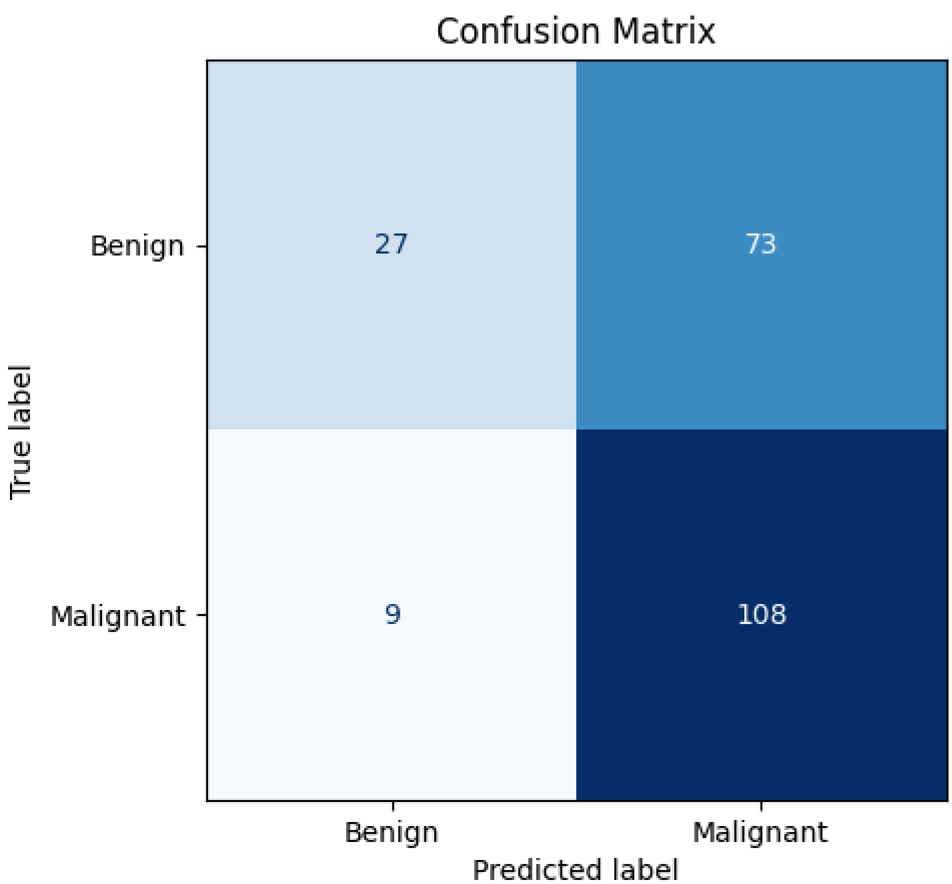


	precision	recall	f1-score	support
Benign	0.69	0.61	0.65	100
Malignant	0.70	0.77	0.73	117
accuracy			0.70	217
macro avg	0.70	0.69	0.69	217
weighted avg	0.70	0.70	0.69	217

40 Epochs

accuracy: 0.6221198439598083,
auc: 0.6786751747131348,
loss: 0.6727909445762634,
precision: 0.5966851115226746,
recall: 0.9230769276618958

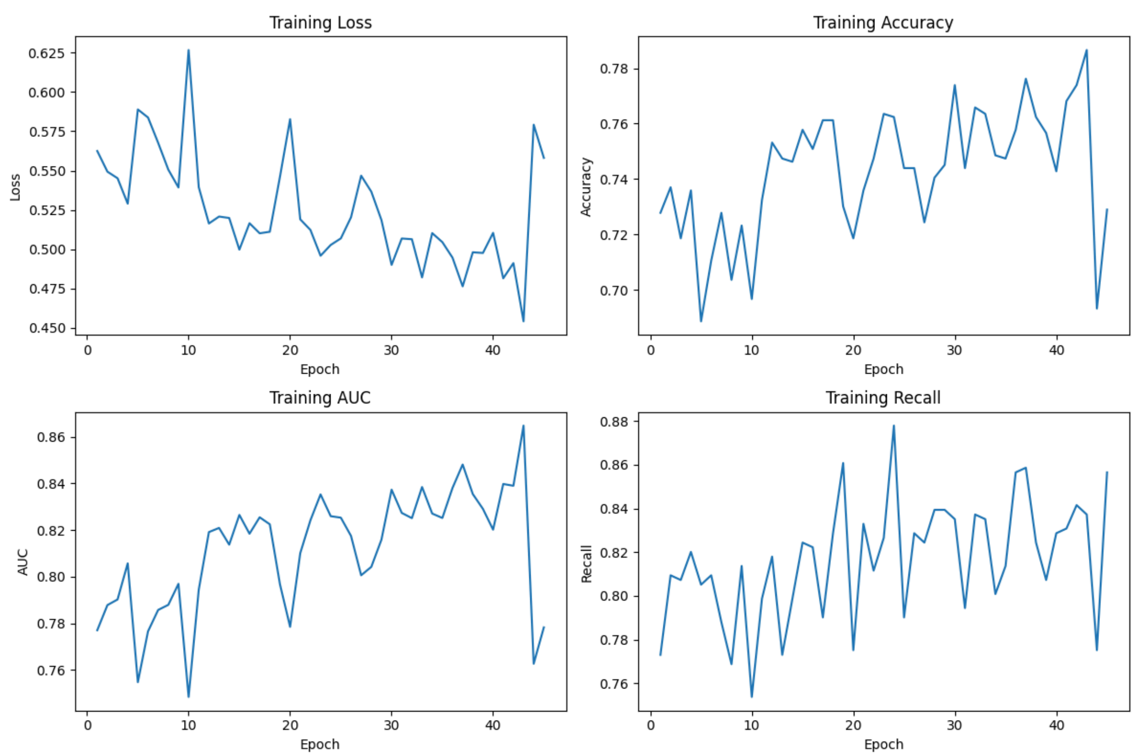

Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch

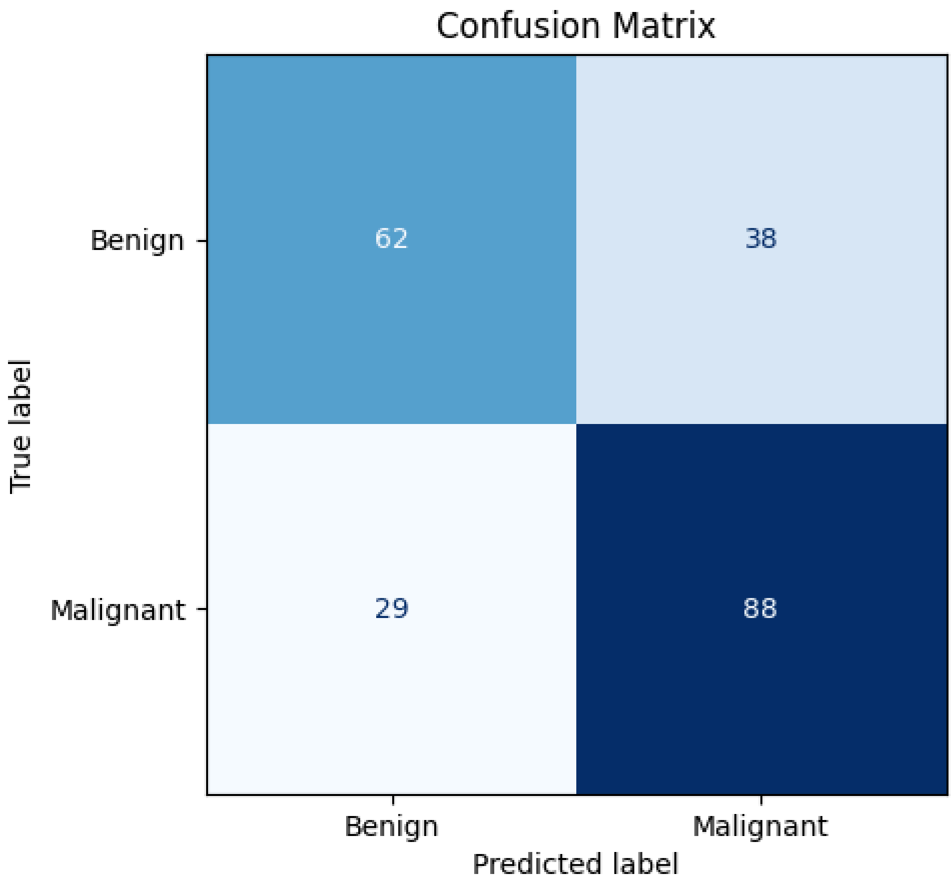

Confusion Matrix
True label
Benign
Malignant
27
73
9
108
Benign
Malignant
Predicted label

	precision	recall	f1-score	support
Benign	0.75	0.27	0.40	100
Malignant	0.60	0.92	0.72	117
accuracy			0.62	217
macro avg	0.67	0.60	0.56	217
weighted avg	0.67	0.62	0.57	217

45 Epochs

accuracy: 0.6912442445755005,
auc: 0.7085469961166382,
loss: 0.8667238354682922,
precision: 0.6984127163887024,
recall: 0.752136766910553

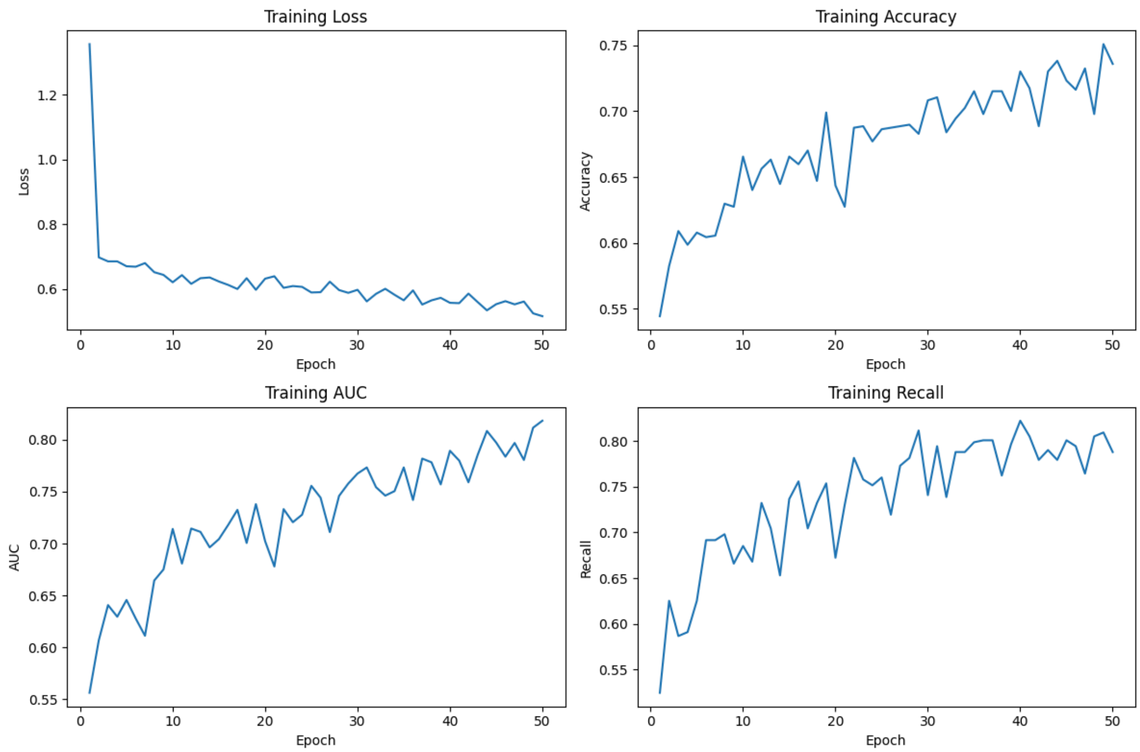




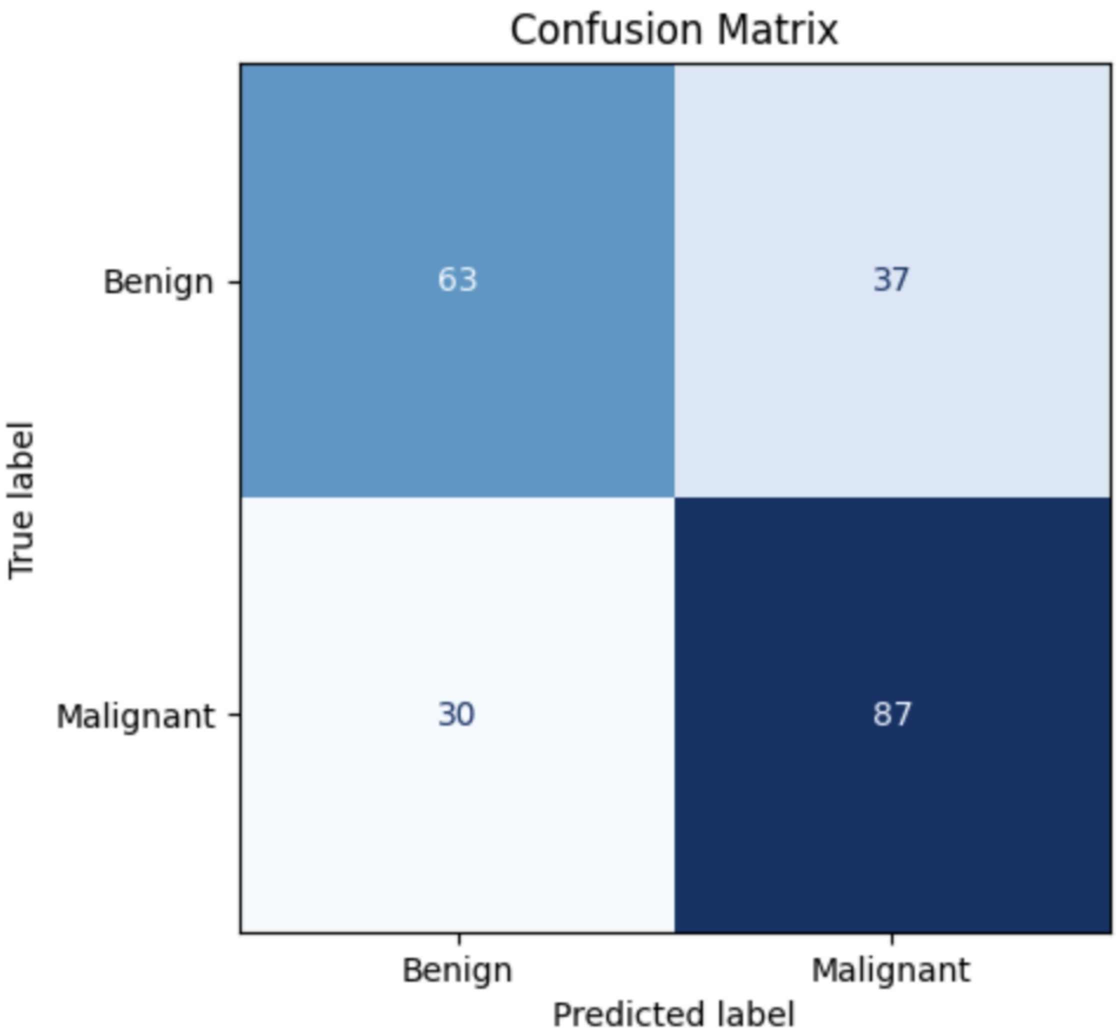

```
              precision    recall  f1-score   support

      Benign       0.68      0.62      0.65       100
   Malignant       0.70      0.75      0.72       117

    accuracy                           0.69       217
   macro avg       0.69      0.69      0.69       217
weighted avg       0.69      0.69      0.69       217
```

50 Epochs

accuracy: 0.6912442445755005,
auc: 0.736709475517273,
loss: 0.6325473785400391,
precision: 0.7016128897666931,
recall': 0.7435897588729858



	precision	recall	f1-score	support
Benign	0.68	0.63	0.65	100
Malignant	0.70	0.74	0.72	117
accuracy			0.69	217
macro avg	0.69	0.69	0.69	217
weighted avg	0.69	0.69	0.69	217



Overall

- accuracy improved to about 71.4%
- malignant recall is still strong: 96 / 117 ≈ 82.1%
- benign recall improved a lot: 59 / 100 = 59%
- malignant precision is about 96 / (96 + 41) ≈ 70.1%

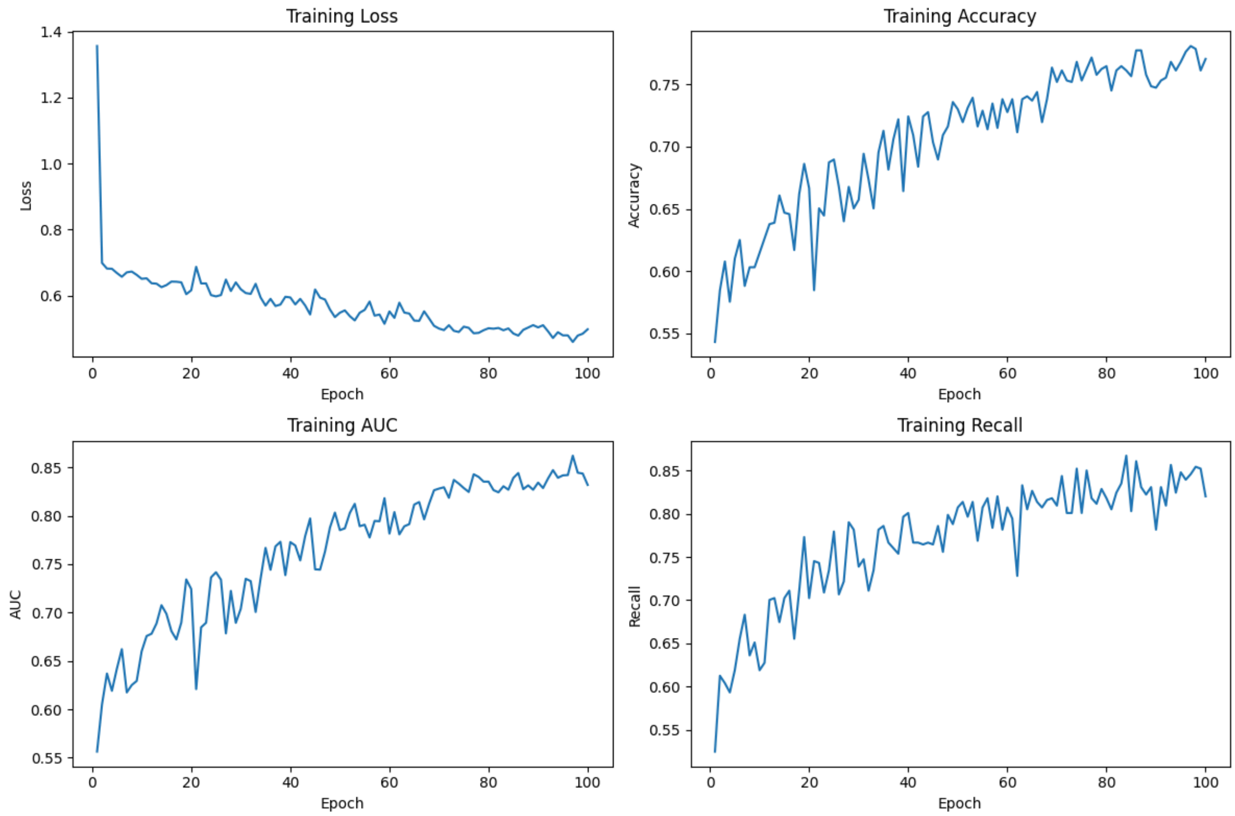
Changed Thresholds

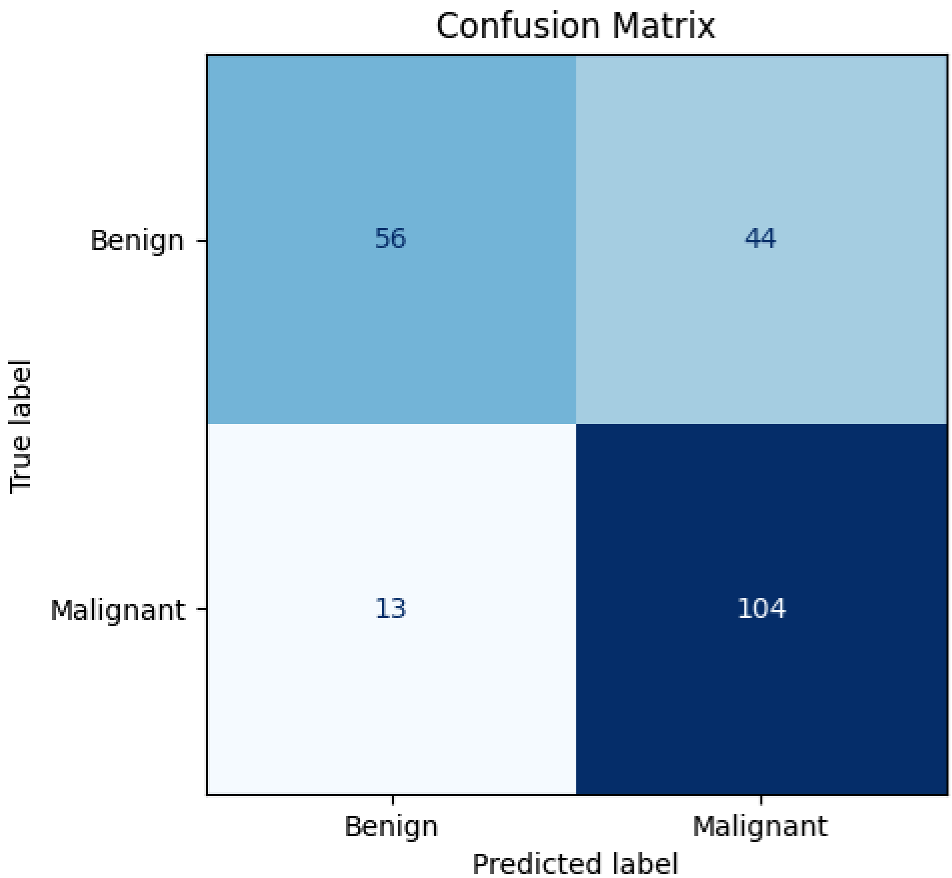
- 0.6 & 0.55: catching way more benign, but missing too many malignant

Threshold of **0.5** was retained as the decision threshold because increasing the threshold improved benign specificity but caused a notable drop in malignant recall, which is less desirable for melanoma screening.

100 Epoch

accuracy: 0.7373272180557251,
auc: 0.7536324262619019,
loss: 0.7799624800682068,
precision: 0.7027027010917664,
recall: 0.8888888955116272

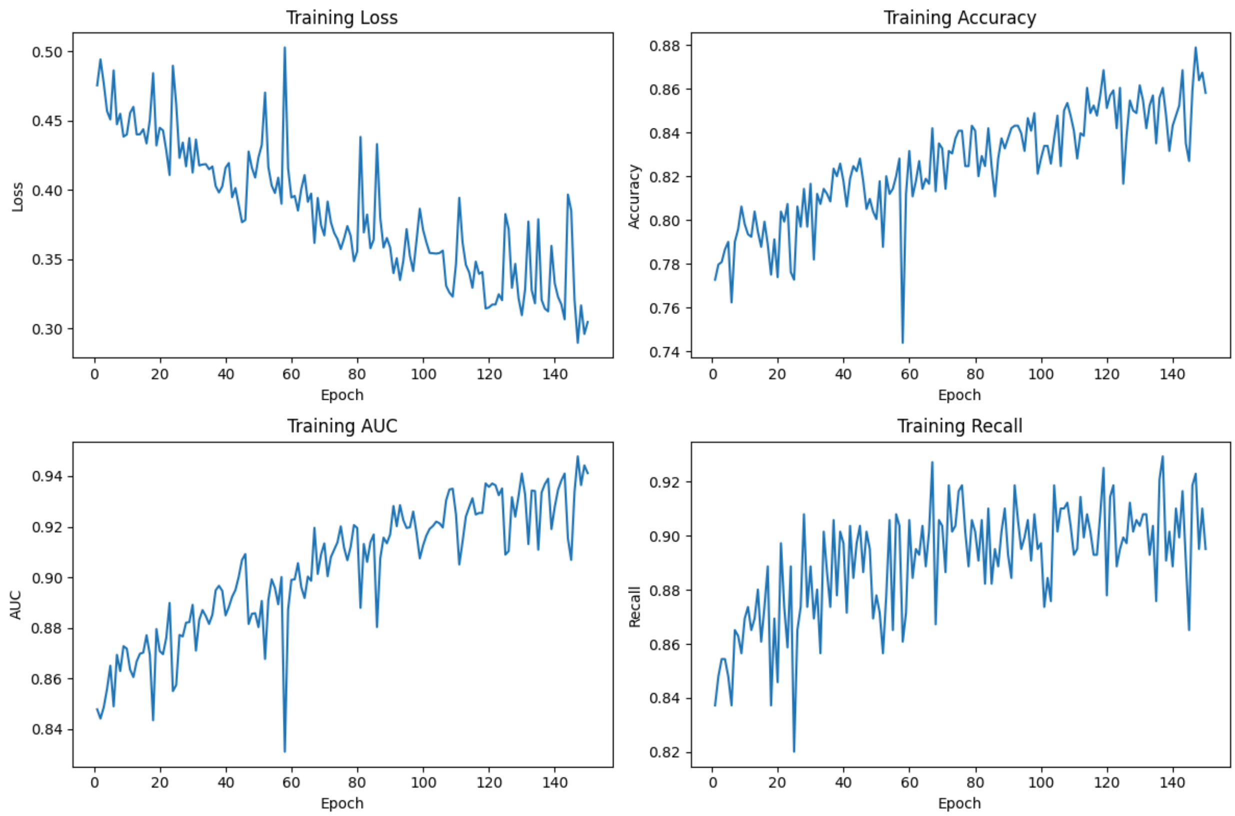




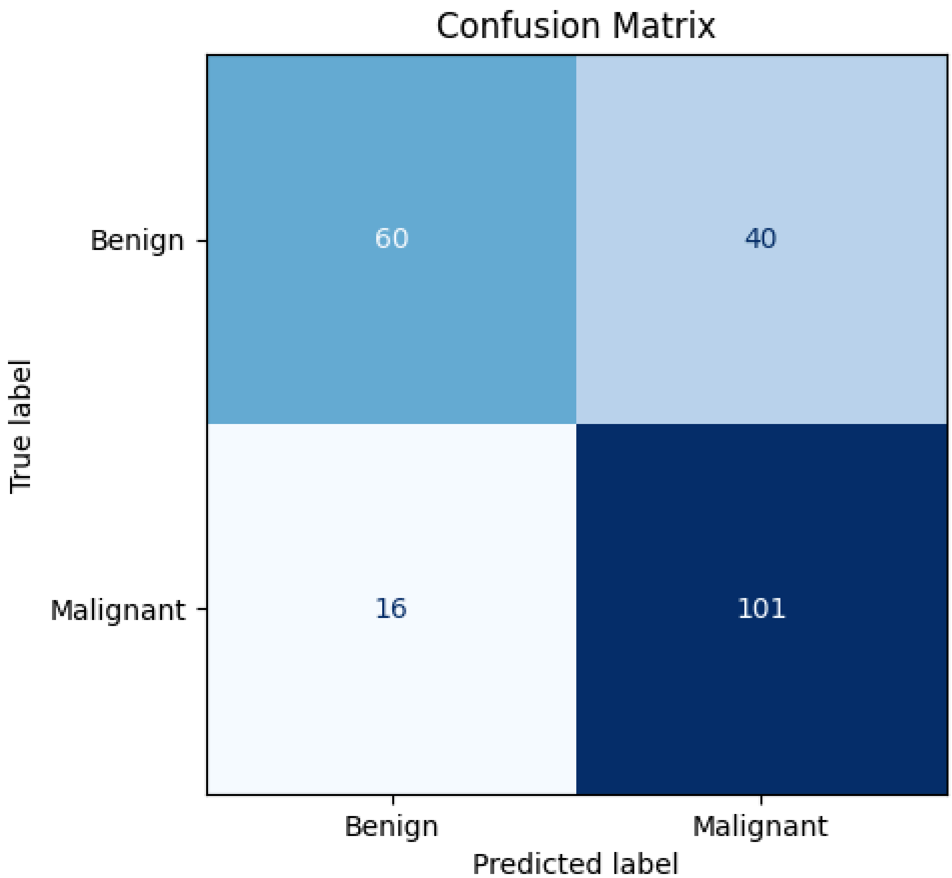
	precision	recall	f1-score	support
Benign	0.81	0.56	0.66	100
Malignant	0.70	0.89	0.78	117
accuracy			0.74	217
macro avg	0.76	0.72	0.72	217
weighted avg	0.75	0.74	0.73	217

150 Epochs

```
{'accuracy': 0.7419354915618896,
 'auc': 0.7780342102050781,
 'loss': 1.2954028844833374,
 'precision': 0.716312050819397,
 'recall': 0.8632478713989258}
```



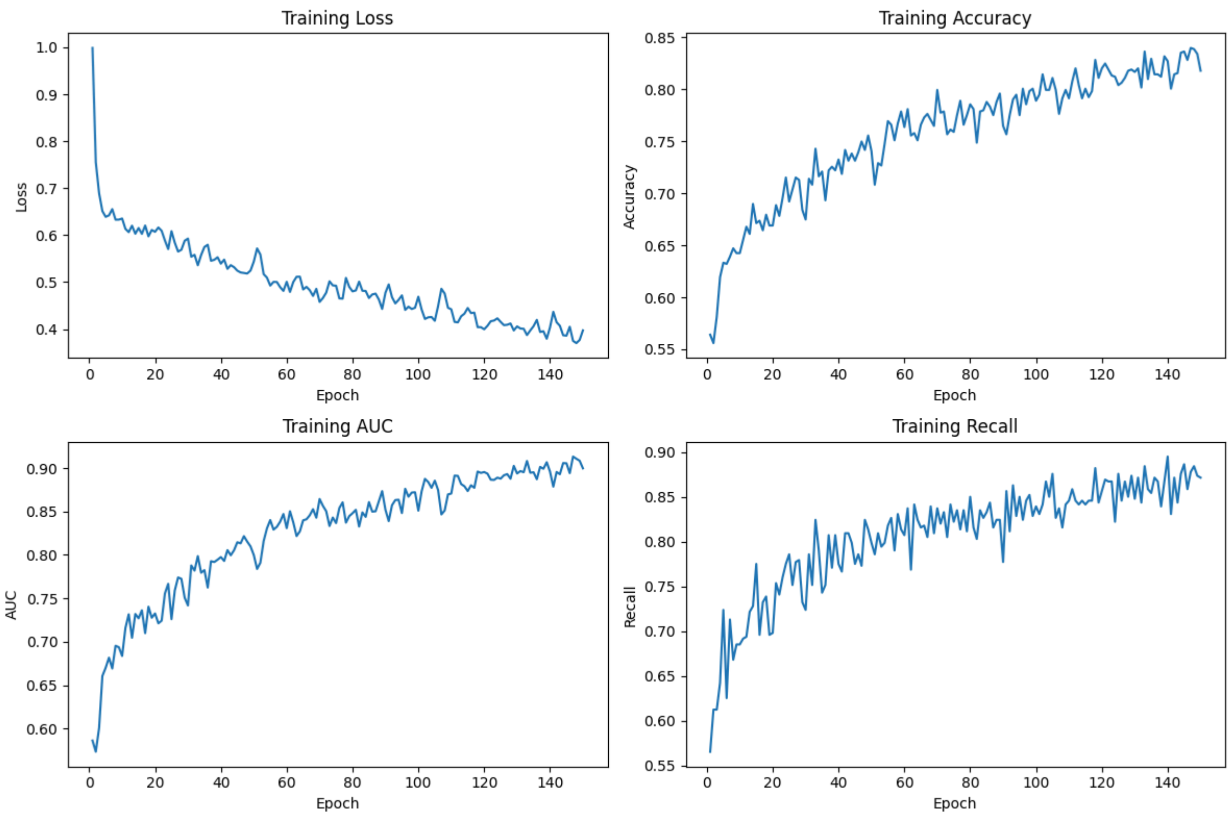
Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch



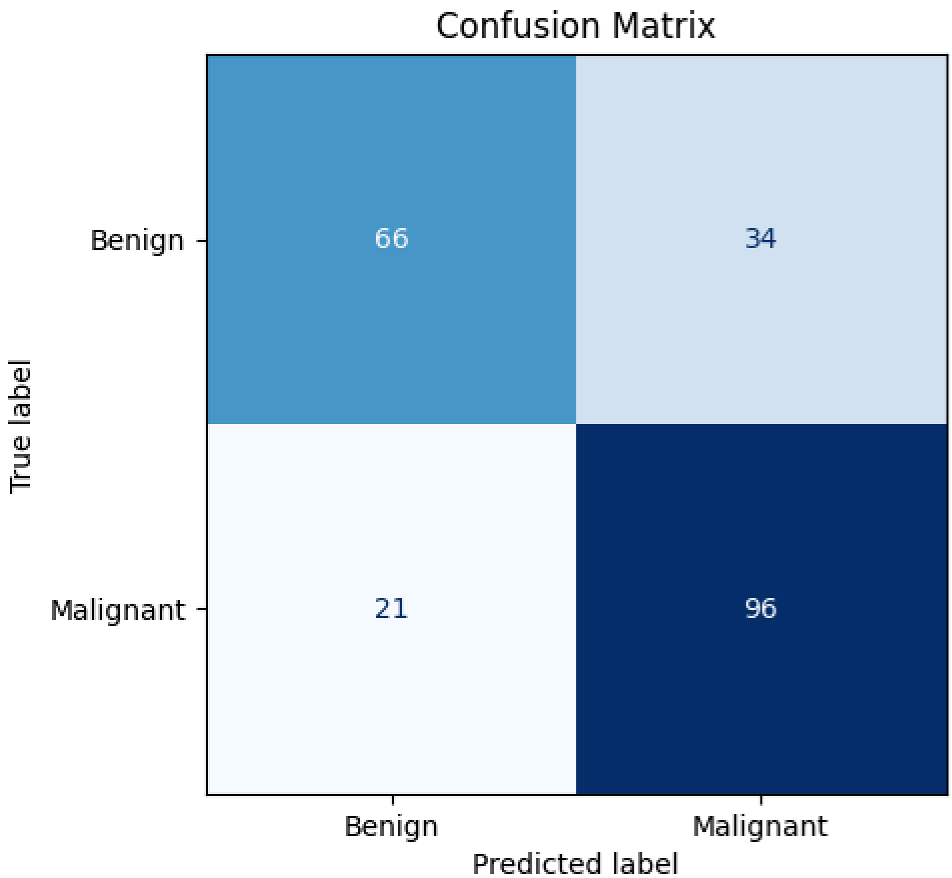
	precision	recall	f1-score	support
Benign	0.79	0.60	0.68	100
Malignant	0.72	0.86	0.78	117
accuracy			0.74	217
macro avg	0.75	0.73	0.73	217
weighted avg	0.75	0.74	0.74	217

64 Batch Size

'accuracy': 0.7465437650680542,
'auc': 0.7892735004425049,
'loss': 0.6735677123069763,
'precision': 0.7384615540504456,
'recall': 0.8205128312110901



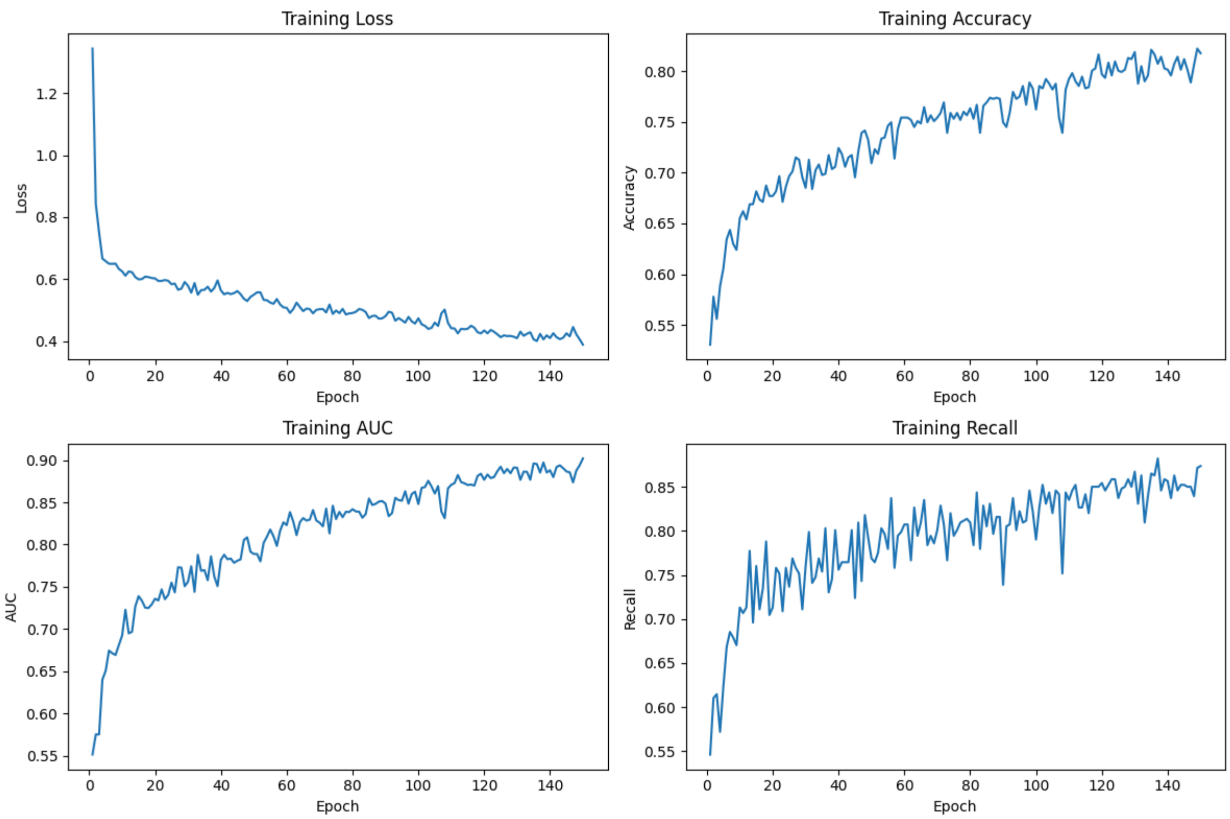
Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch

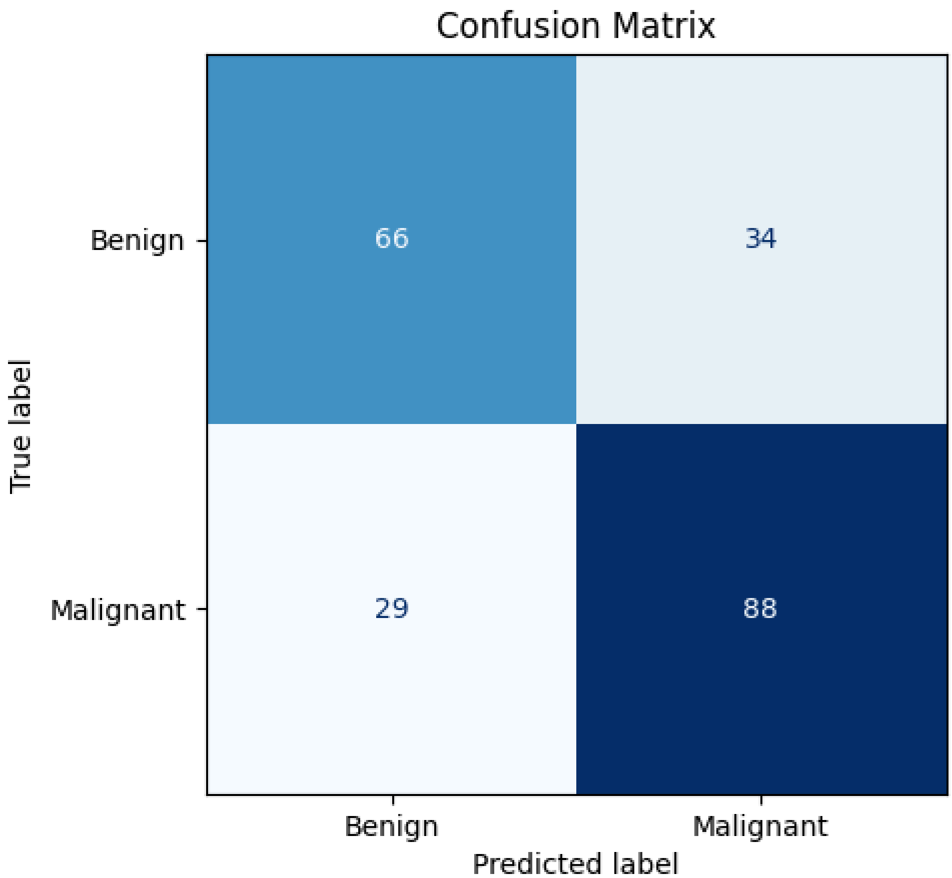


	precision	recall	f1-score	support
Benign	0.76	0.66	0.71	100
Malignant	0.74	0.82	0.78	117
accuracy			0.75	217
macro avg	0.75	0.74	0.74	217
weighted avg	0.75	0.75	0.74	217

64 Batch Size & 0.0005 Learning Rate

'accuracy': 0.7096773982048035,
'auc': 0.7766666412353516,
'loss': 0.6294366717338562,
'precision': 0.7213114500045776,
'recall': 0.752136766910553

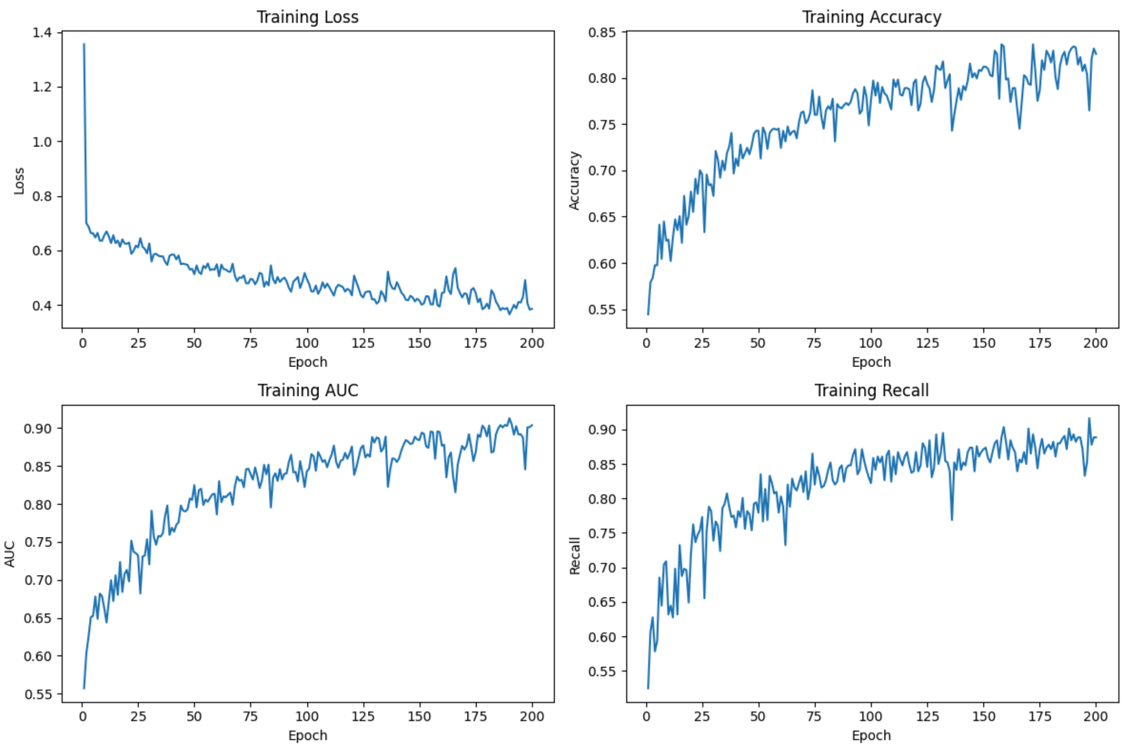

Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch

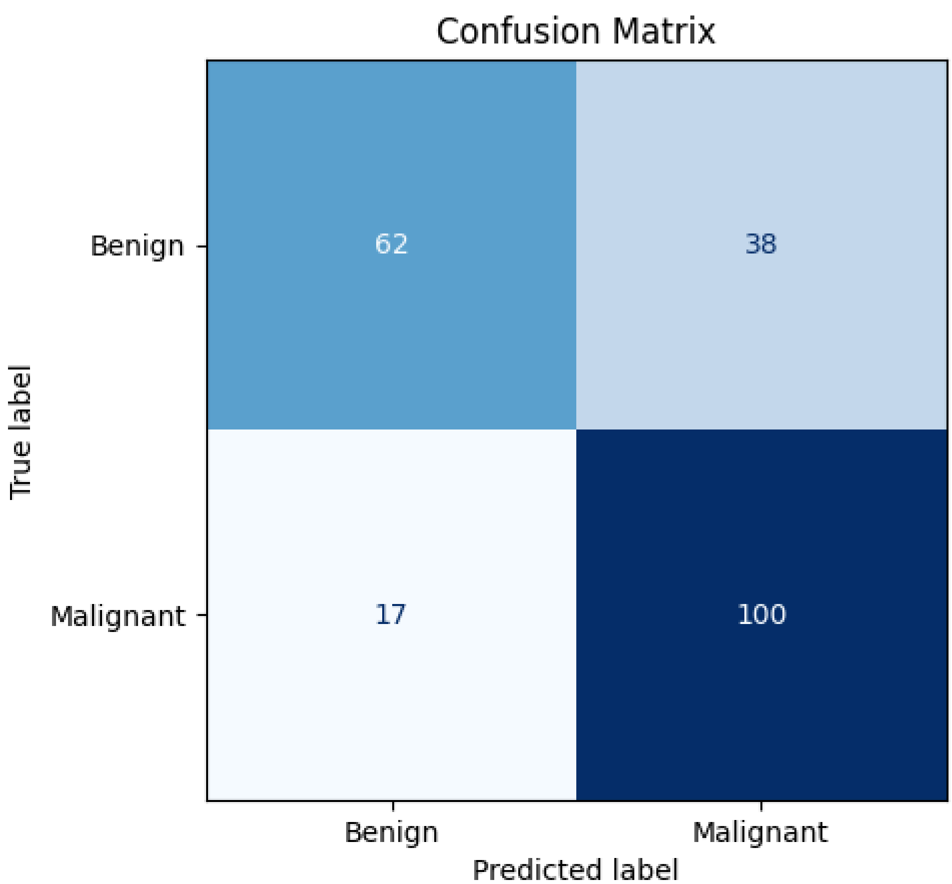


	precision	recall	f1-score	support
Benign	0.69	0.66	0.68	100
Malignant	0.72	0.75	0.74	117
accuracy			0.71	217
macro avg	0.71	0.71	0.71	217
weighted avg	0.71	0.71	0.71	217

200 Epochs

{'accuracy': 0.7465437650680542,
'auc': 0.7677350640296936,
'loss': 0.9530345797538757,
'precision': 0.7246376872062683,
'recall': 0.8547008633613586}

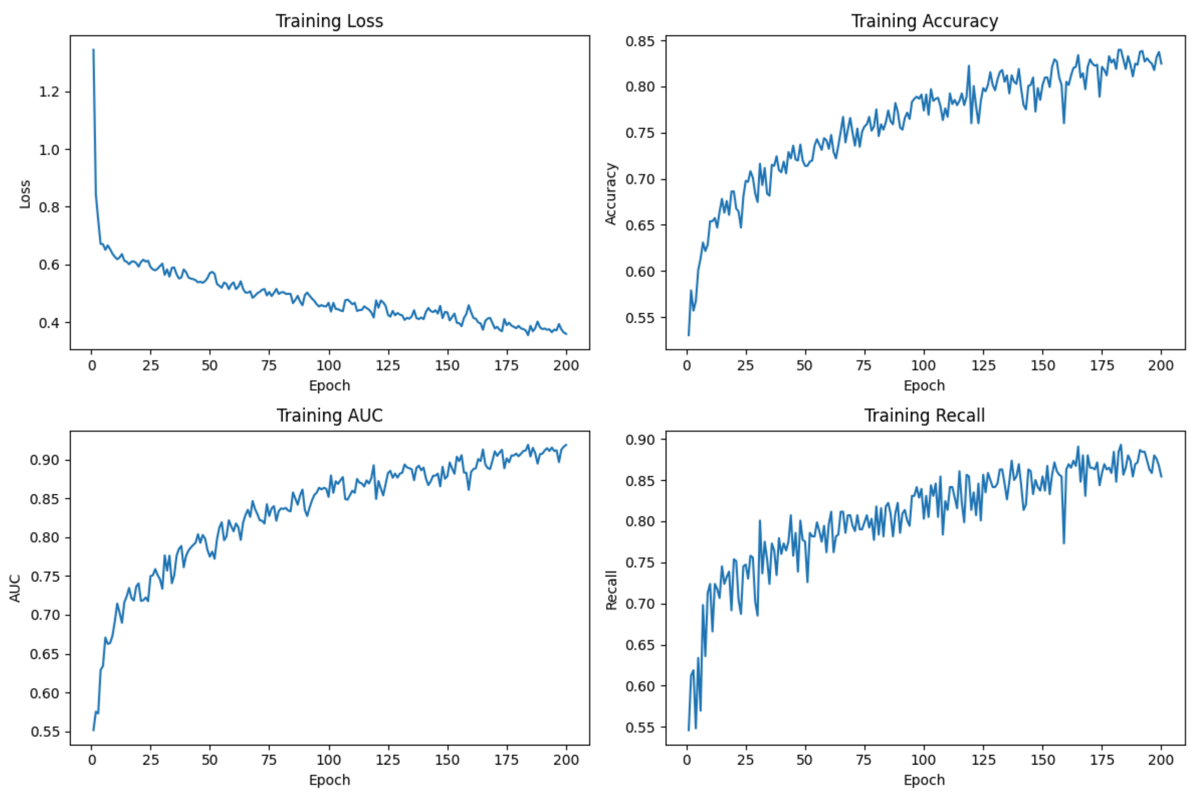

Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch

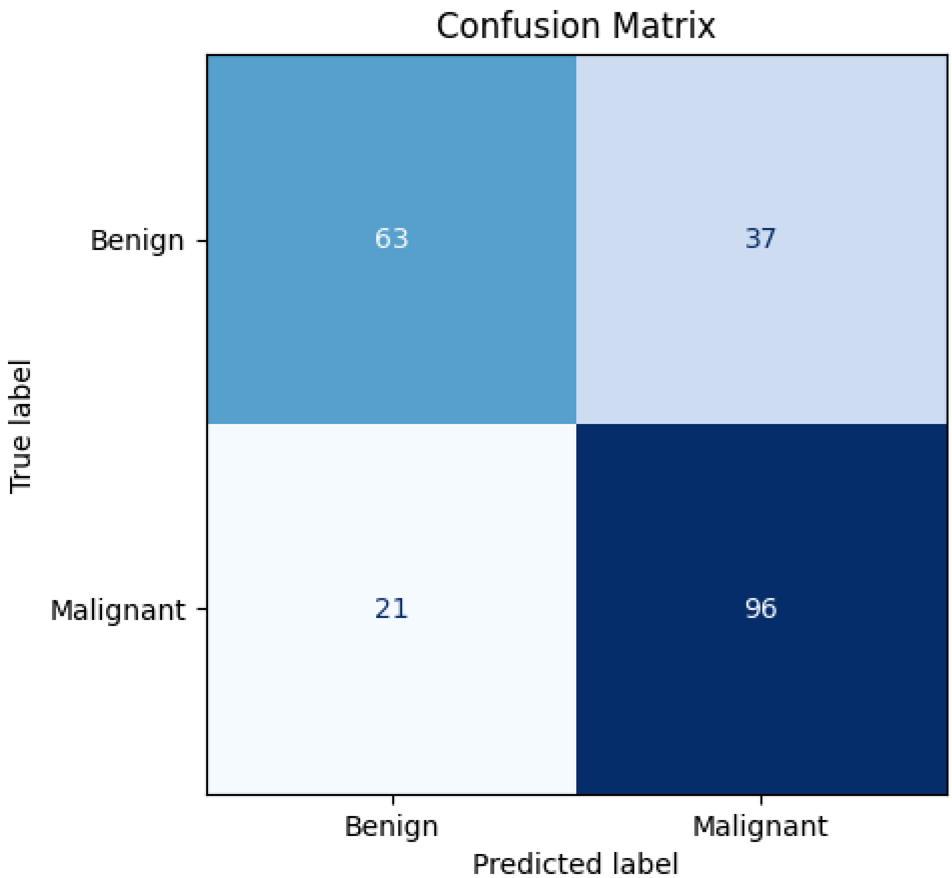

Confusion Matrix
Benign
Malignant
True label
62
38
17
100
Benign
Malignant
Predicted label

	precision	recall	f1-score	support
Benign	0.78	0.62	0.69	100
Malignant	0.72	0.85	0.78	117
accuracy			0.75	217
macro avg	0.75	0.74	0.74	217
weighted avg	0.75	0.75	0.74	217

64 Batch Size & 0.0005 Learning

'accuracy': 0.7327188849449158,
'auc': 0.7861111760139465,
'loss': 0.7371823787689209,
'precision': 0.7218044996261597,
'recall': 0.8205128312110901





	precision	recall	f1-score	support
Benign	0.75	0.63	0.68	100
Malignant	0.72	0.82	0.77	117
accuracy			0.73	217
macro avg	0.74	0.73	0.73	217
weighted avg	0.73	0.73	0.73	217

Overall

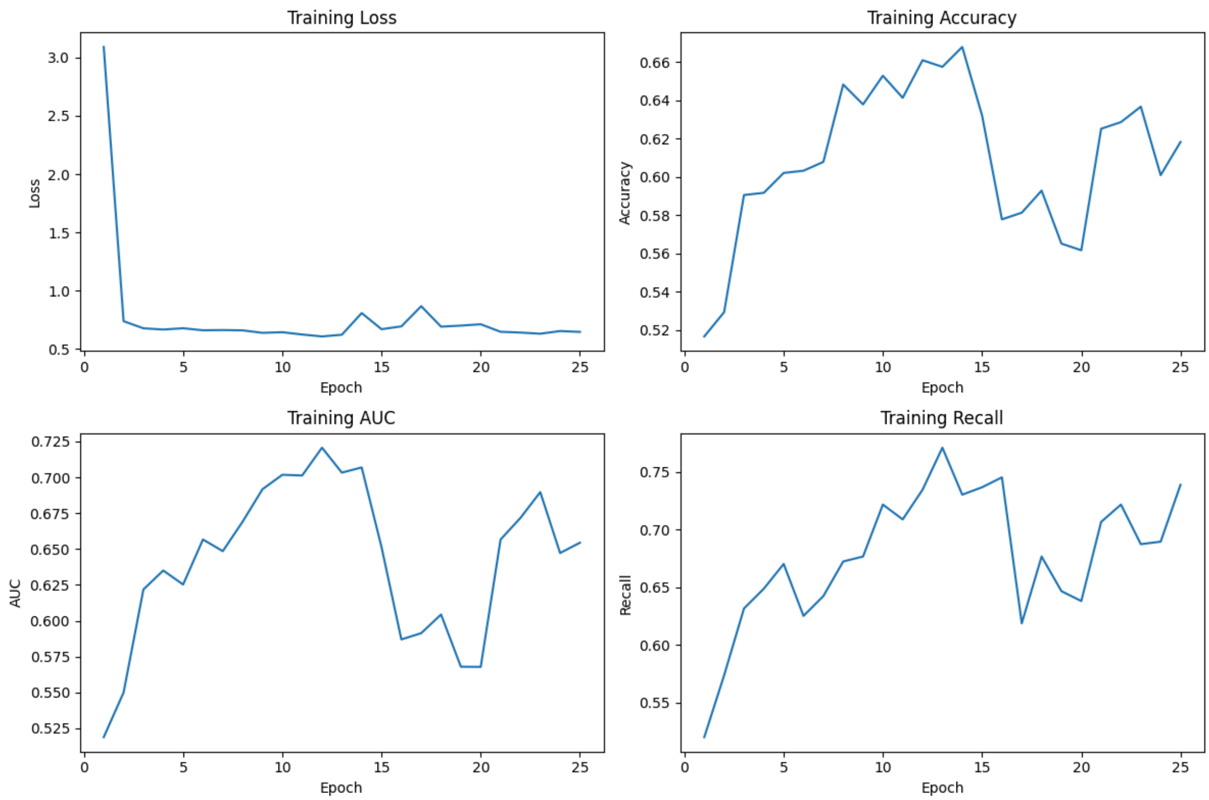
Best performing is **35 epochs** standard **threshold of 0.5**

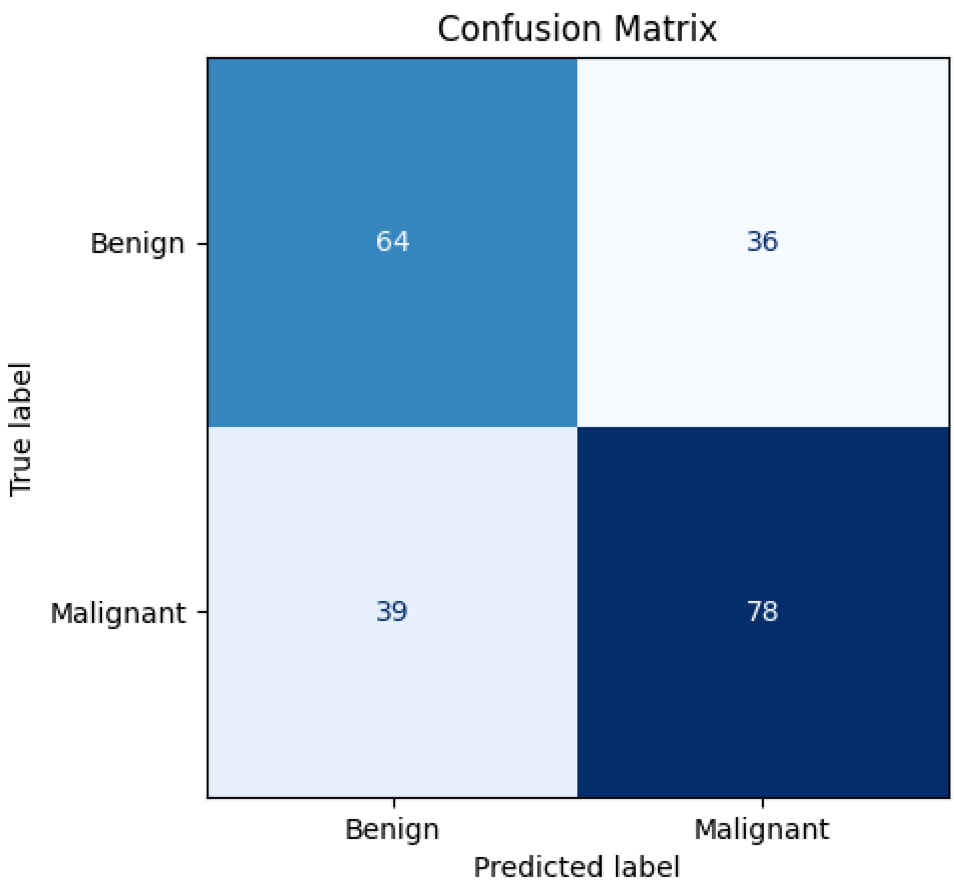
Model Architecture 2:

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_1 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_2 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 128)	0
dense (Dense)	(None, 128)	16,512
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 1)	129

25 Epochs

accuracy: 0.6497696042060852
auc: 0.6937606930732727,
loss: 0.6565001606941223,
precision: 0.6171428561210632,
recall: 0.9230769276618958

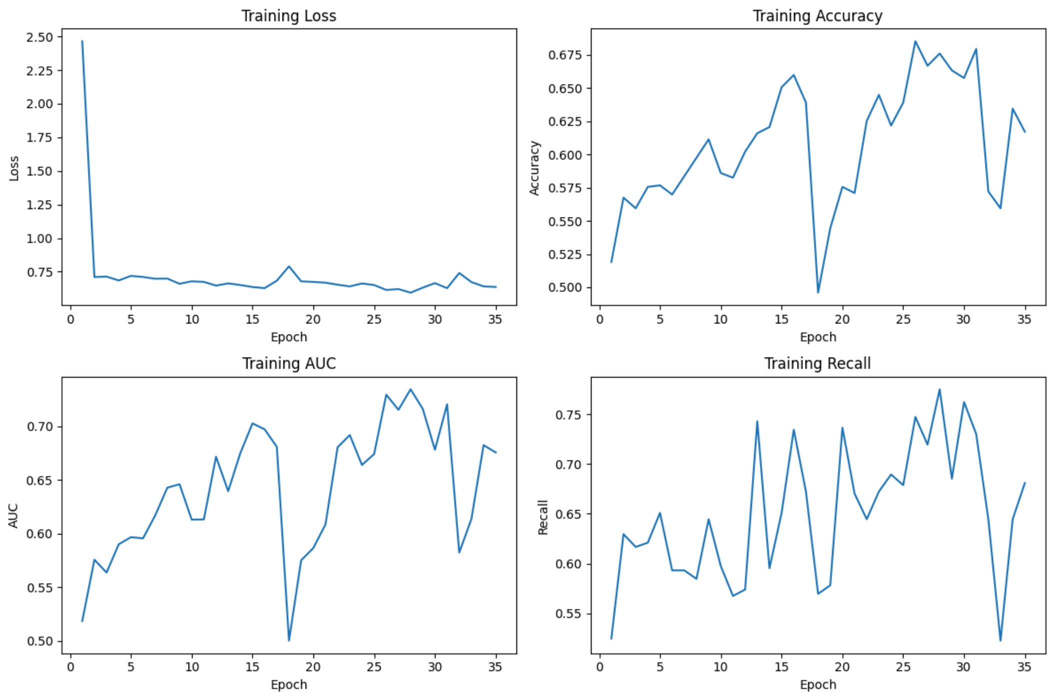

Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch



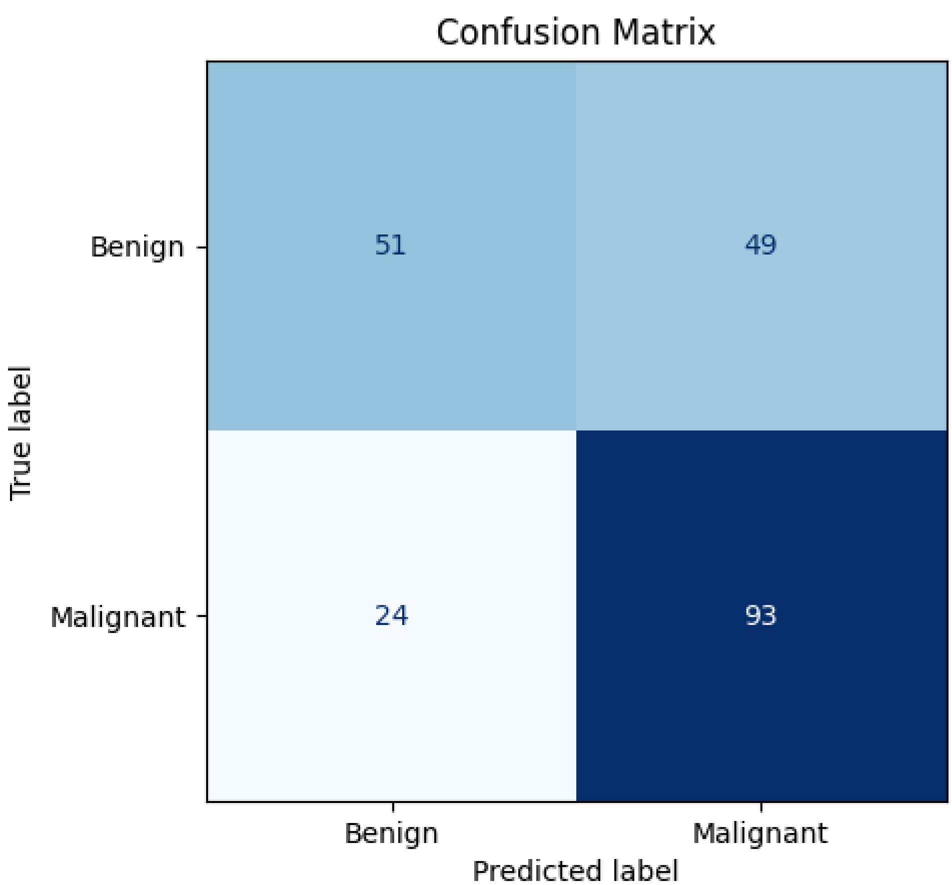
	precision	recall	f1-score	support
Benign	0.62	0.64	0.63	100
Malignant	0.68	0.67	0.68	117
accuracy			0.65	217
macro avg	0.65	0.65	0.65	217
weighted avg	0.66	0.65	0.65	217

35 Epochs

accuracy: 0.6635944843292236,
auc: 0.7362393736839294,
loss: 0.6435627937316895,
precision: 0.6549295783042908,
recall: 0.7948718070983887



Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch

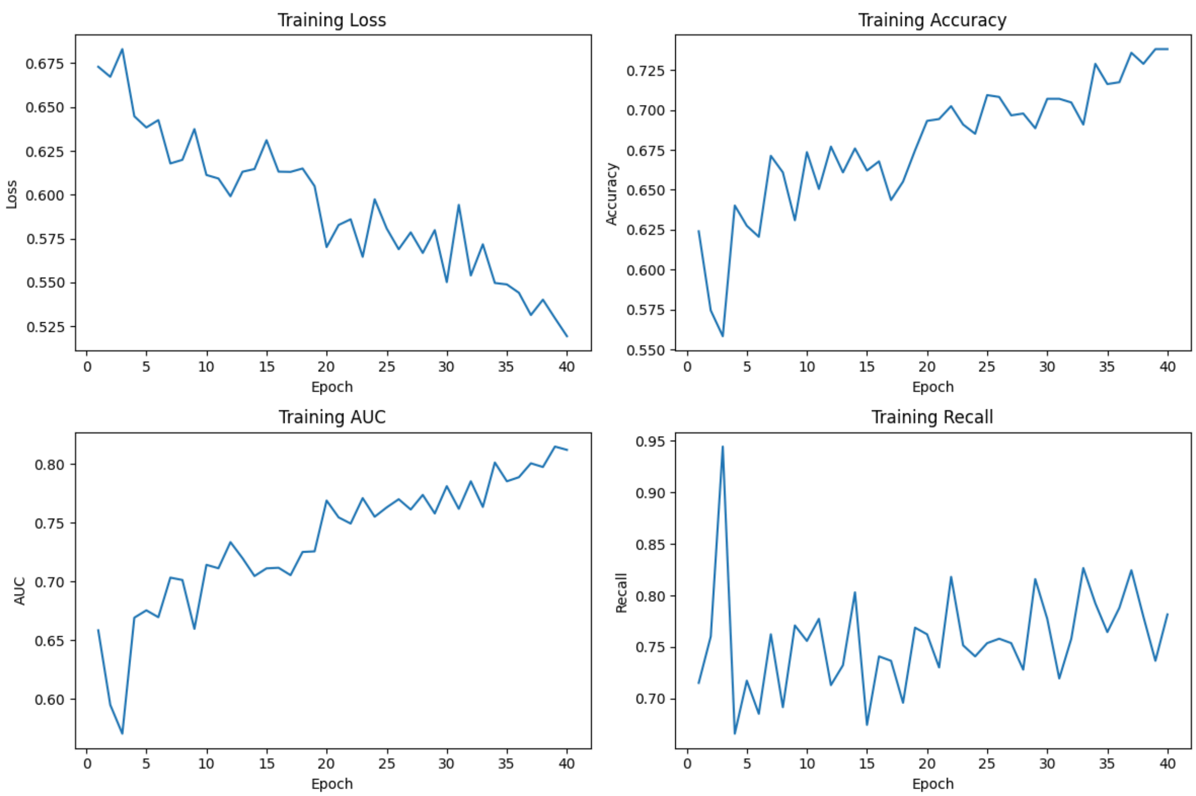


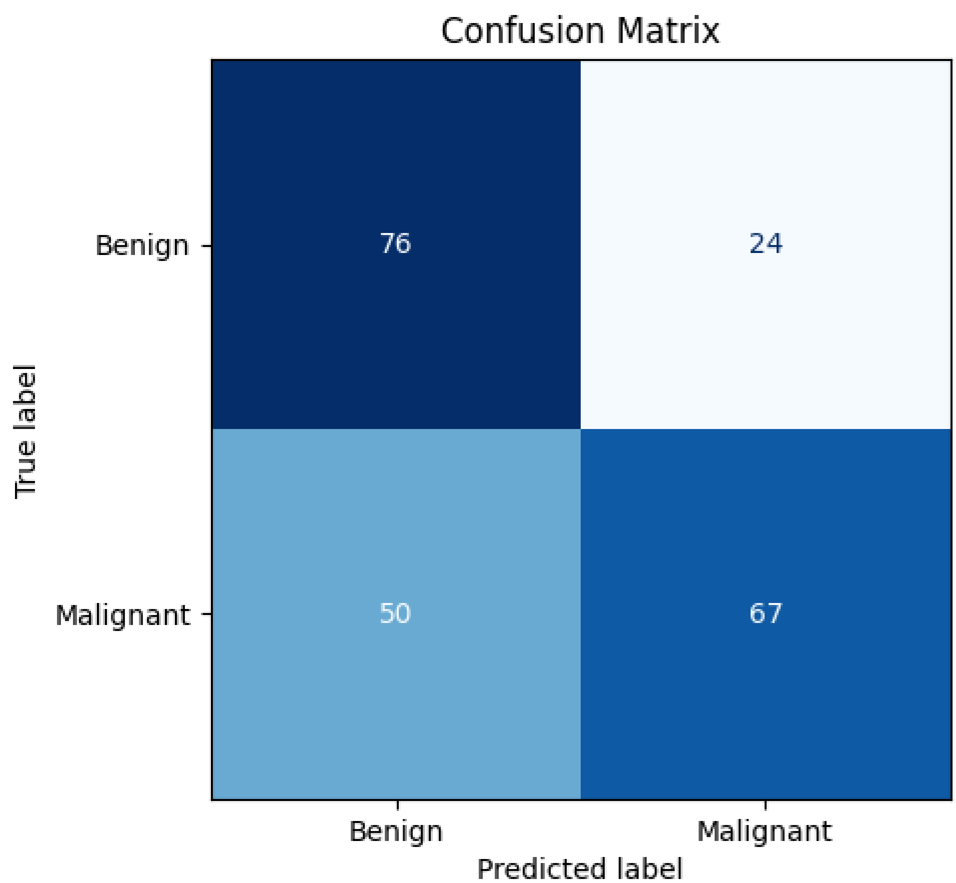
Confusion Matrix
True label
Benign
Malignant
51
49
24
93
Benign
Malignant
Predicted label

	precision	recall	f1-score	support
Benign	0.68	0.51	0.58	100
Malignant	0.65	0.79	0.72	117
accuracy			0.66	217
macro avg	0.67	0.65	0.65	217
weighted avg	0.67	0.66	0.66	217

40 Epochs

accuracy: 0.6589861512184143,
auc: 0.7617949843406677,
loss: 0.6837283968925476,
precision: 0.7362637519836426,
recall: 0.5726495981216431



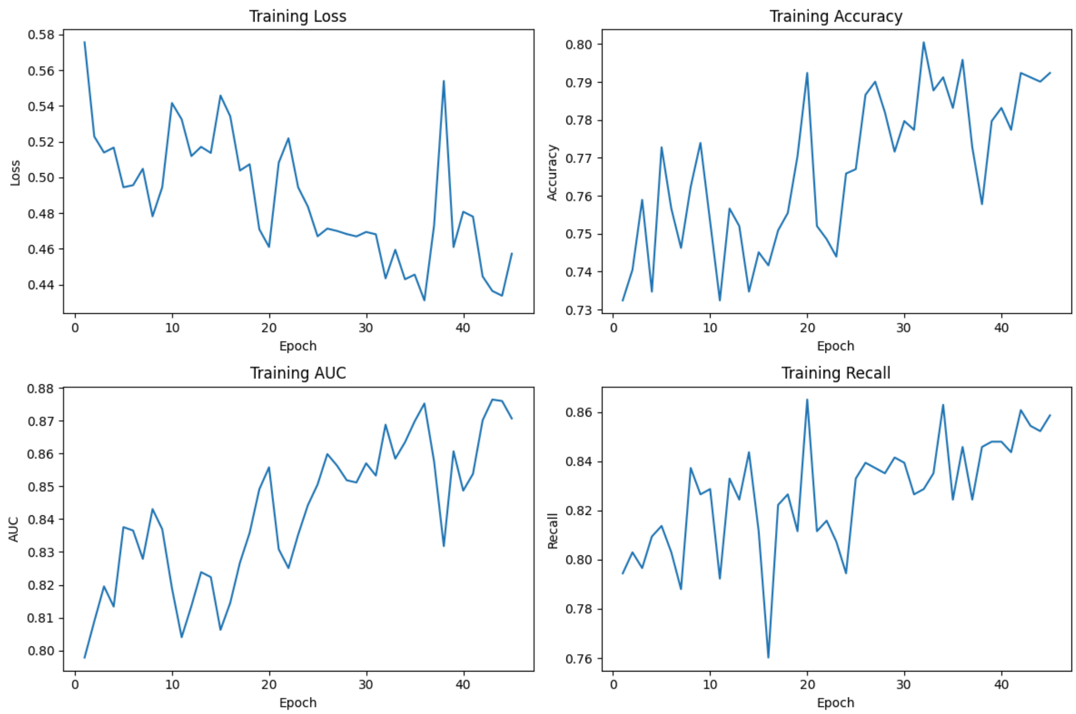


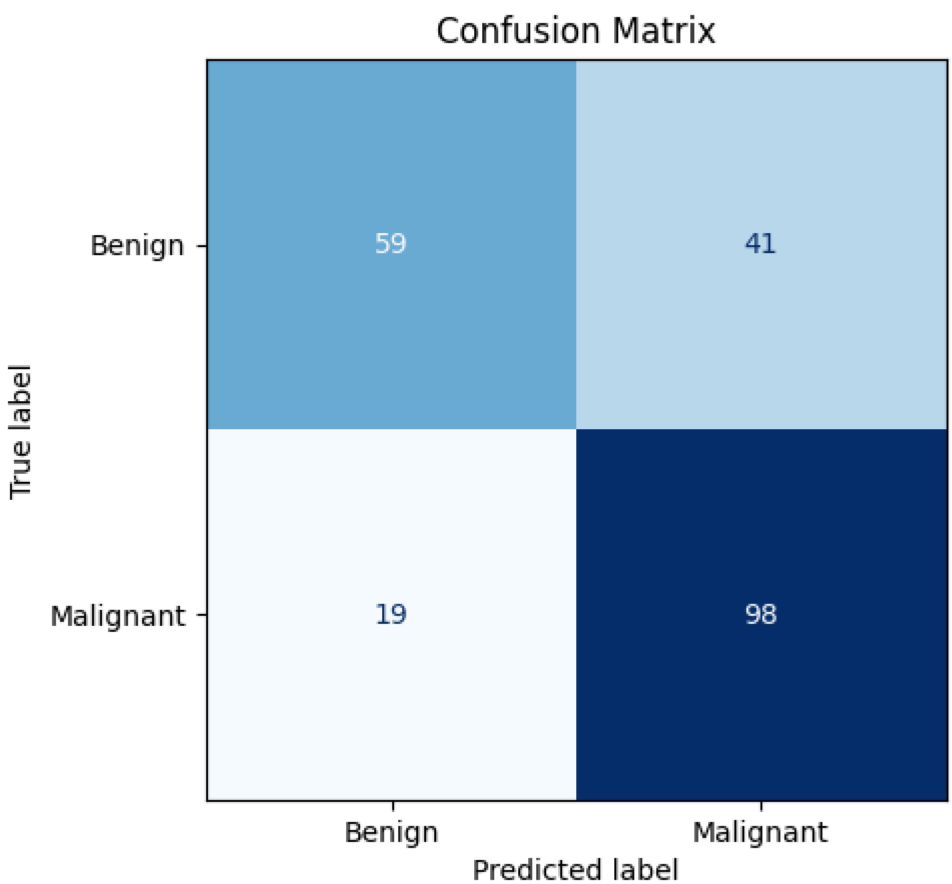
	precision	recall	f1-score	support
Benign	0.60	0.76	0.67	100
Malignant	0.74	0.57	0.64	117
accuracy			0.66	217
macro avg	0.67	0.67	0.66	217
weighted avg	0.67	0.66	0.66	217

45 Epochs

accuracy: 0.7235022783279419,
auc: 0.78166663646698,
loss: 0.6178155541419983,
precision: 0.7050359845161438,

recall: 0.8376068472862244

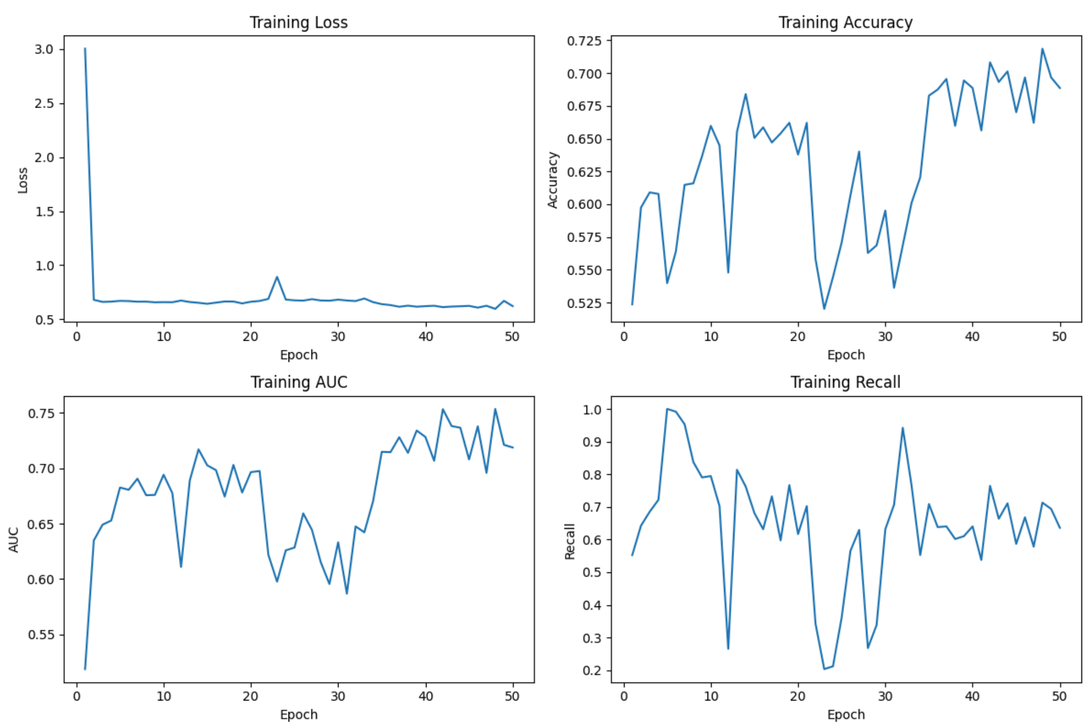


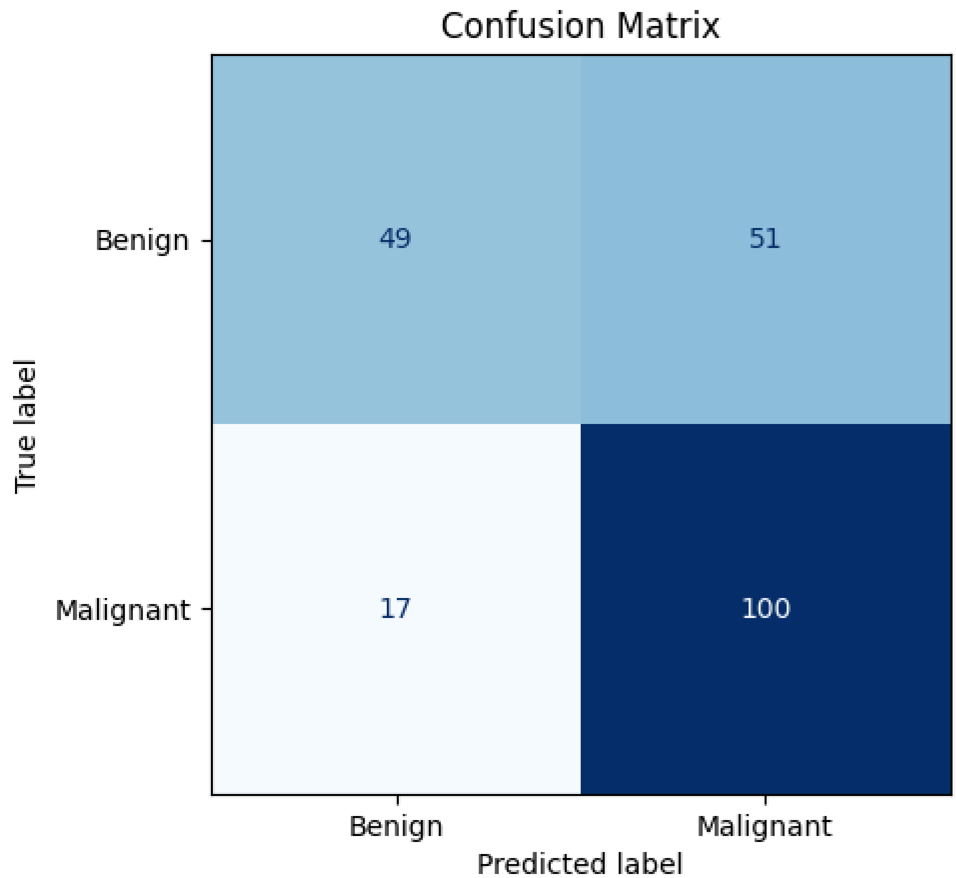


	precision	recall	f1-score	support
Benign	0.76	0.59	0.66	100
Malignant	0.71	0.84	0.77	117
accuracy			0.72	217
macro avg	0.73	0.71	0.71	217
weighted avg	0.73	0.72	0.72	217

50 Epochs

accuracy: 0.6866359710693359,
auc: 0.6675640940666199,
loss: 0.6624300479888916,
precision: 0.6622516512870789,
recall: 0.8547008633613586

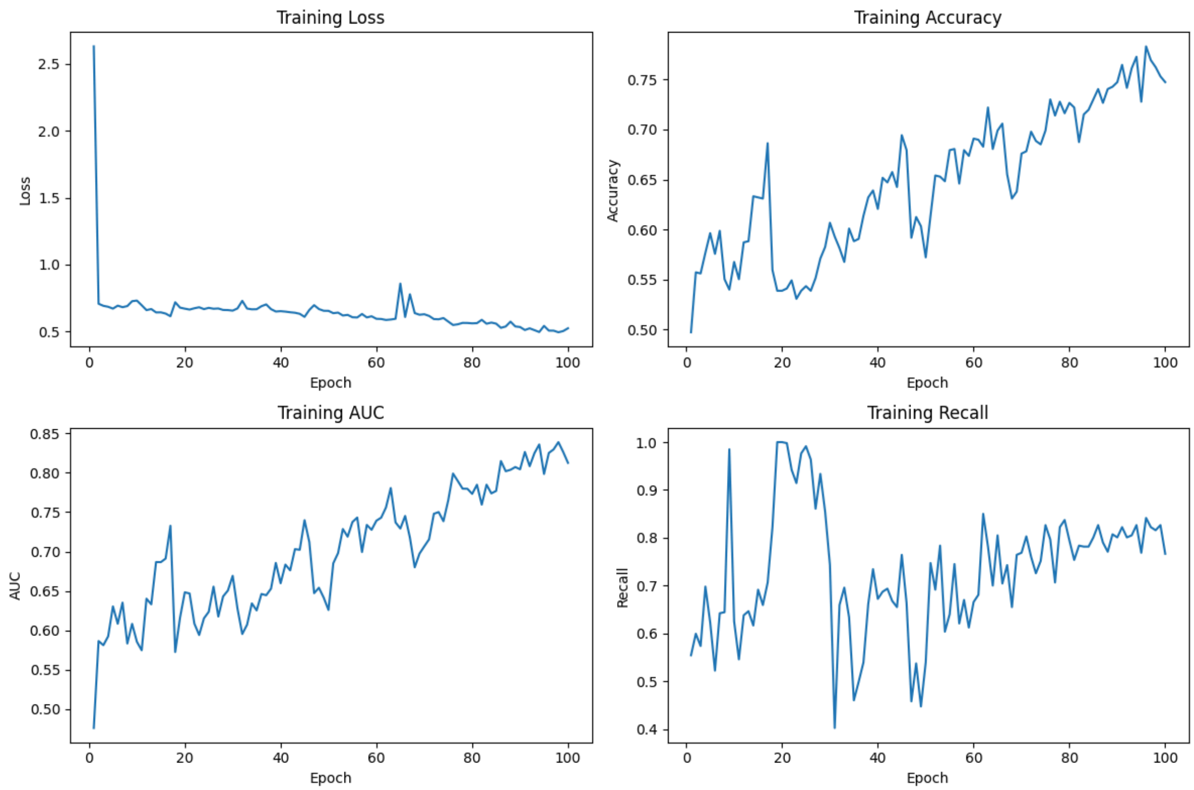


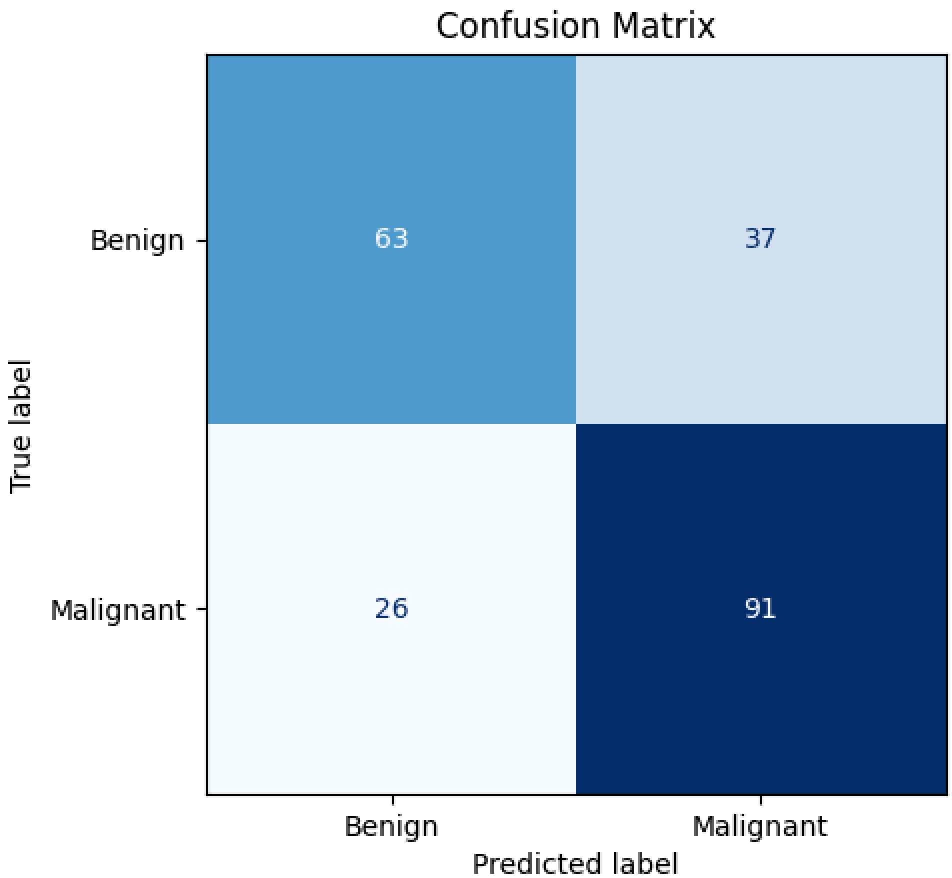


	precision	recall	f1-score	support
Benign	0.74	0.49	0.59	100
Malignant	0.66	0.85	0.75	117
accuracy			0.69	217
macro avg	0.70	0.67	0.67	217
weighted avg	0.70	0.69	0.67	217

100 Epochs

{'accuracy': 0.7096773982048035,
'auc': 0.7398717403411865,
'loss': 1.1786770820617676,
'precision': 0.7109375,
'recall': 0.7777777910232544}

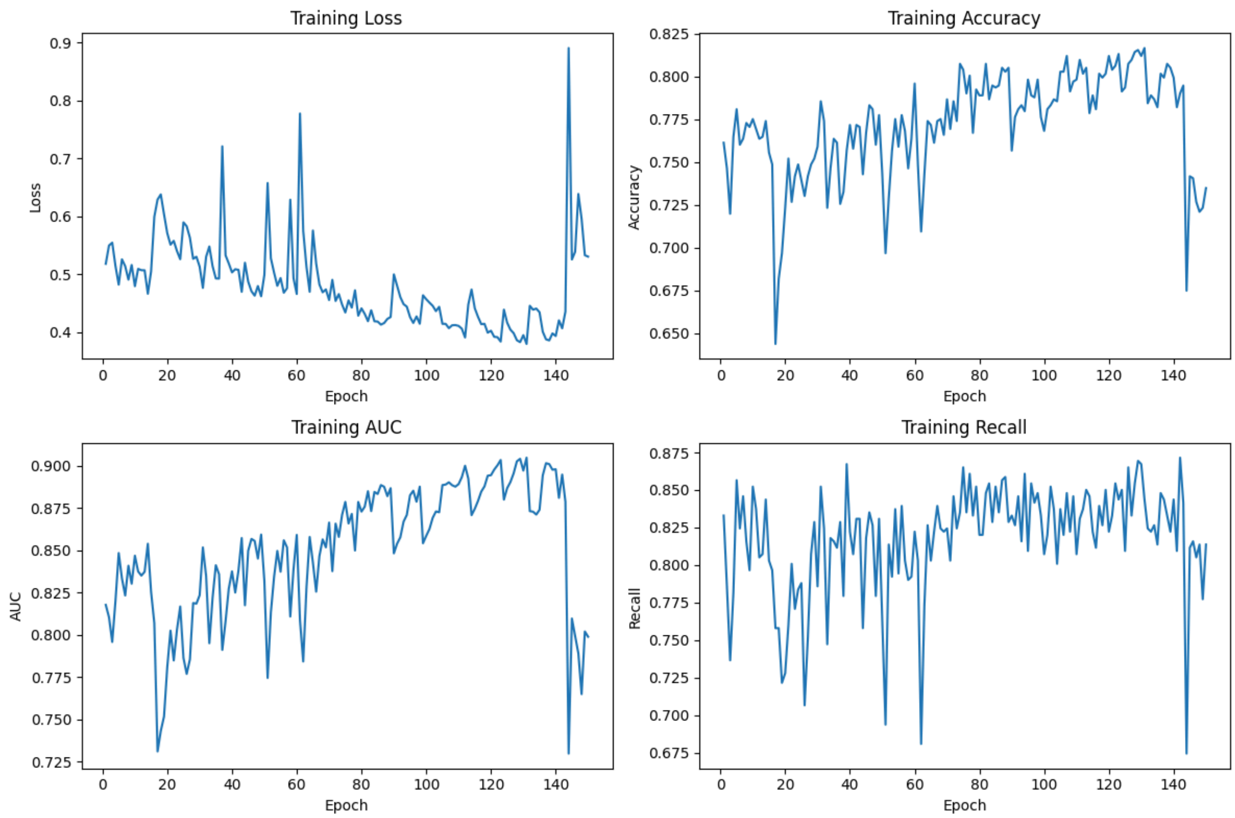

Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch

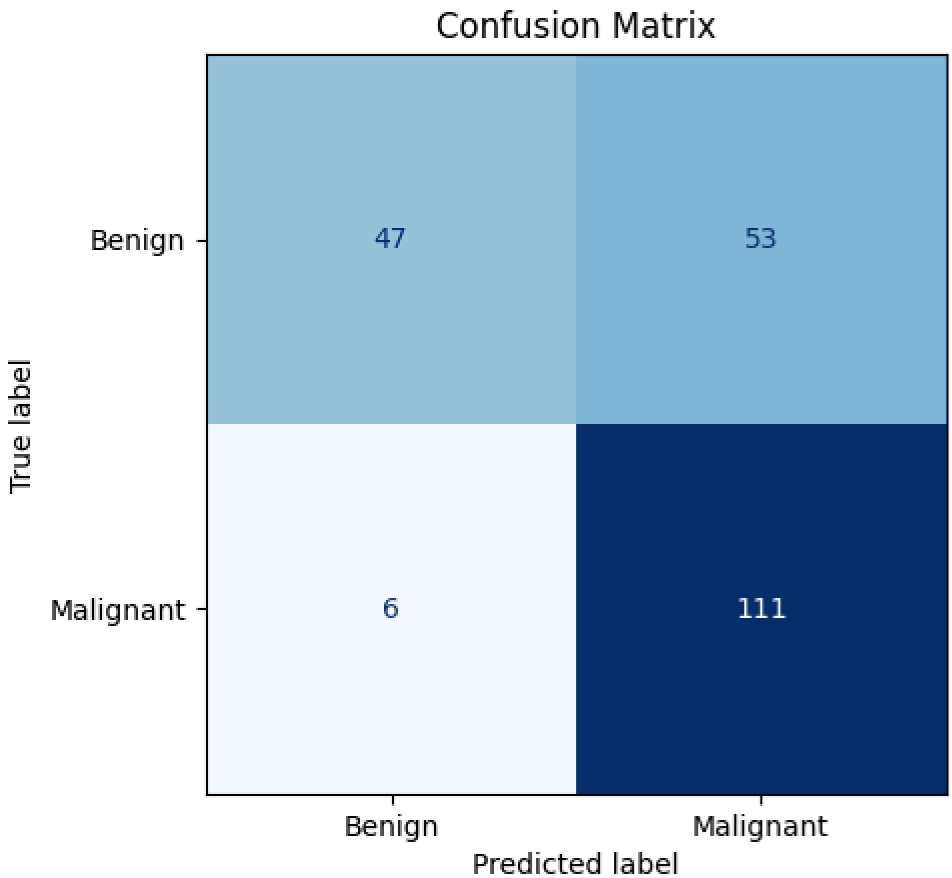


	precision	recall	f1-score	support
Benign	0.71	0.63	0.67	100
Malignant	0.71	0.78	0.74	117
accuracy			0.71	217
macro avg	0.71	0.70	0.70	217
weighted avg	0.71	0.71	0.71	217

150 Epochs

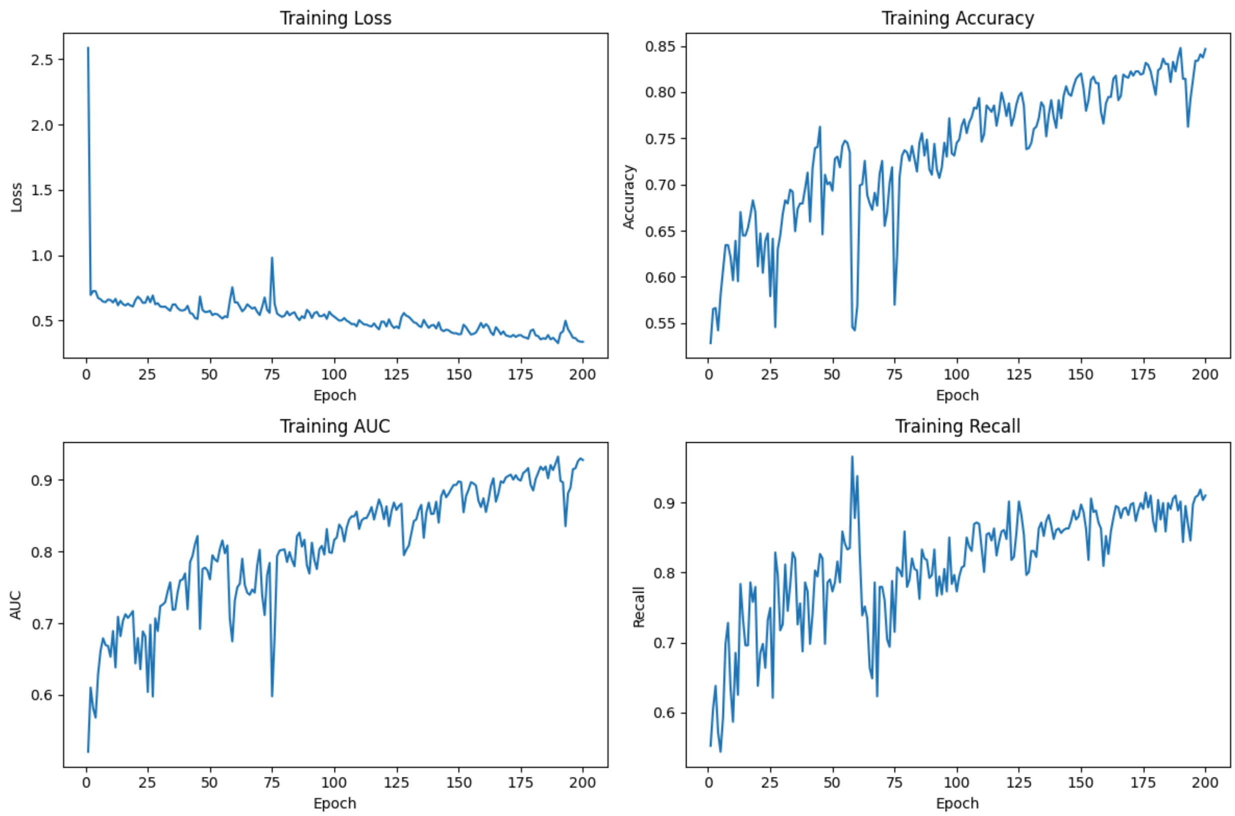
```
{'accuracy': 0.7281106114387512,
 'auc': 0.7723076939582825,
 'loss': 0.576659619808197,
 'precision': 0.6768292784690857,
 'recall': 0.9487179517745972}
```

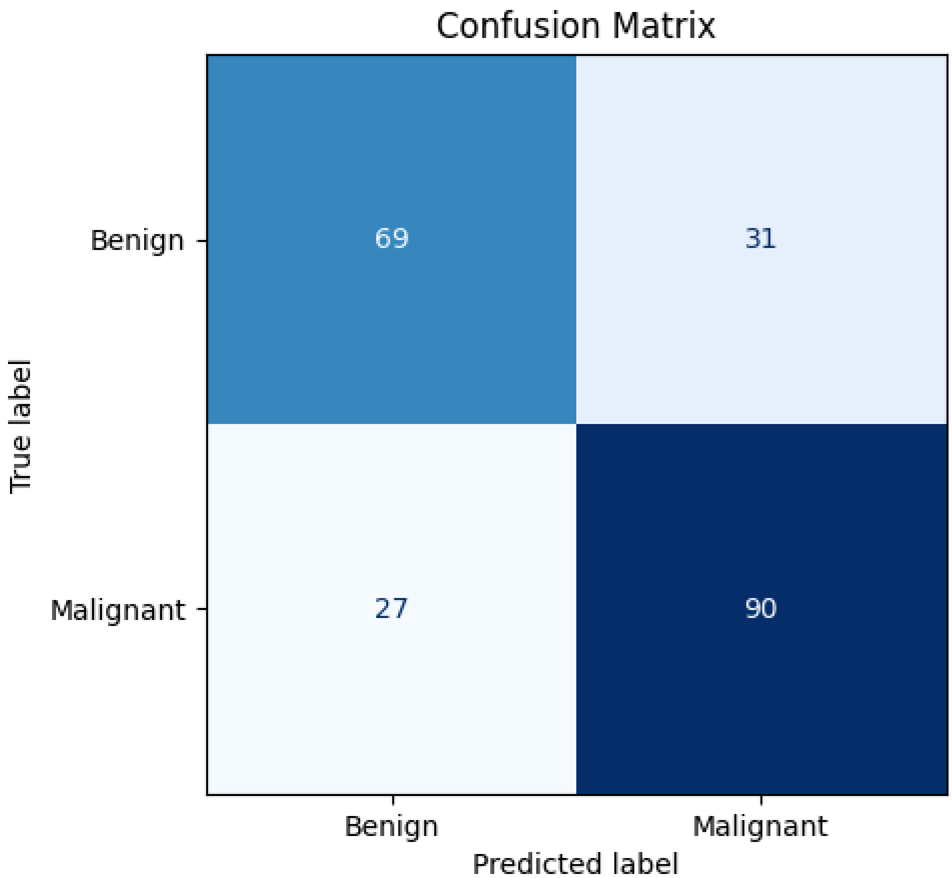

Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch



200 Epochs

{'accuracy': 0.7327188849449158,
'auc': 0.7859401702880859,
'loss': 2.5303590297698975,
'precision': 0.7438016533851624,
'recall': 0.7692307829856873}


Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch



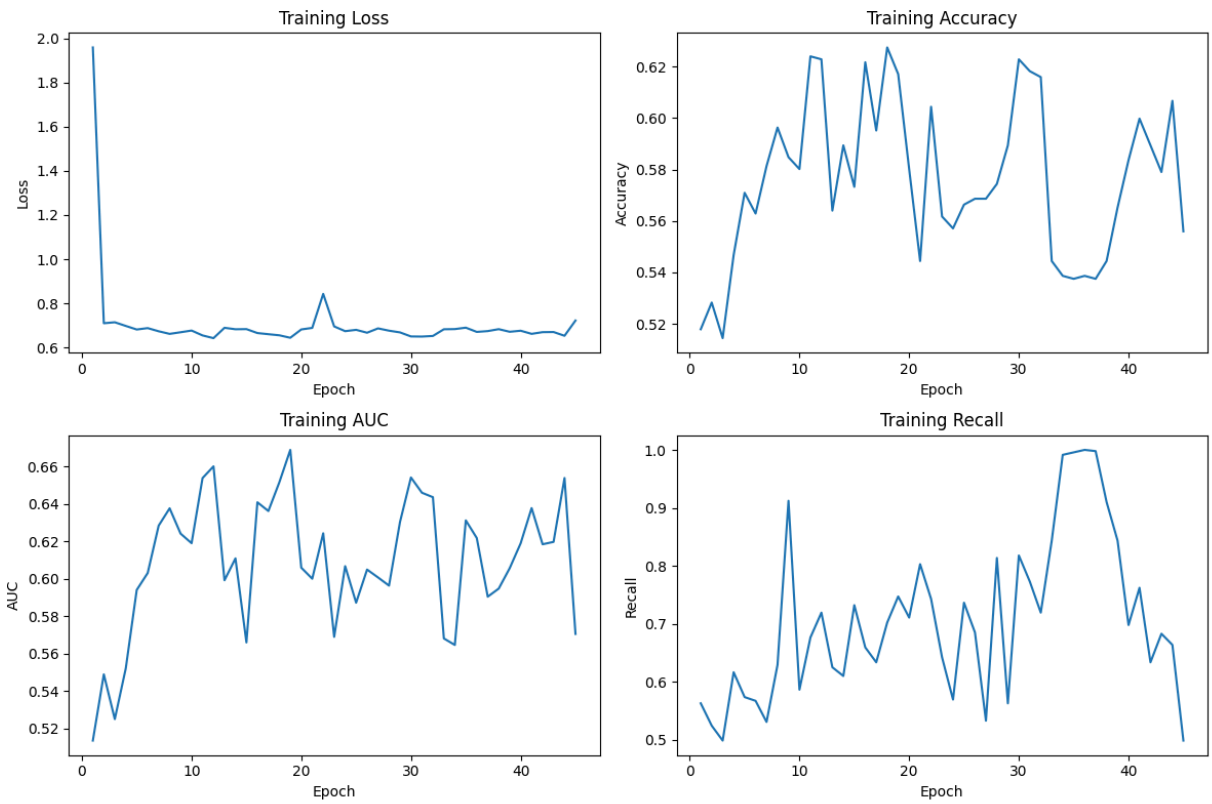
	precision	recall	f1-score	support
Benign	0.72	0.69	0.70	100
Malignant	0.74	0.77	0.76	117
accuracy			0.73	217
macro avg	0.73	0.73	0.73	217
weighted avg	0.73	0.73	0.73	217

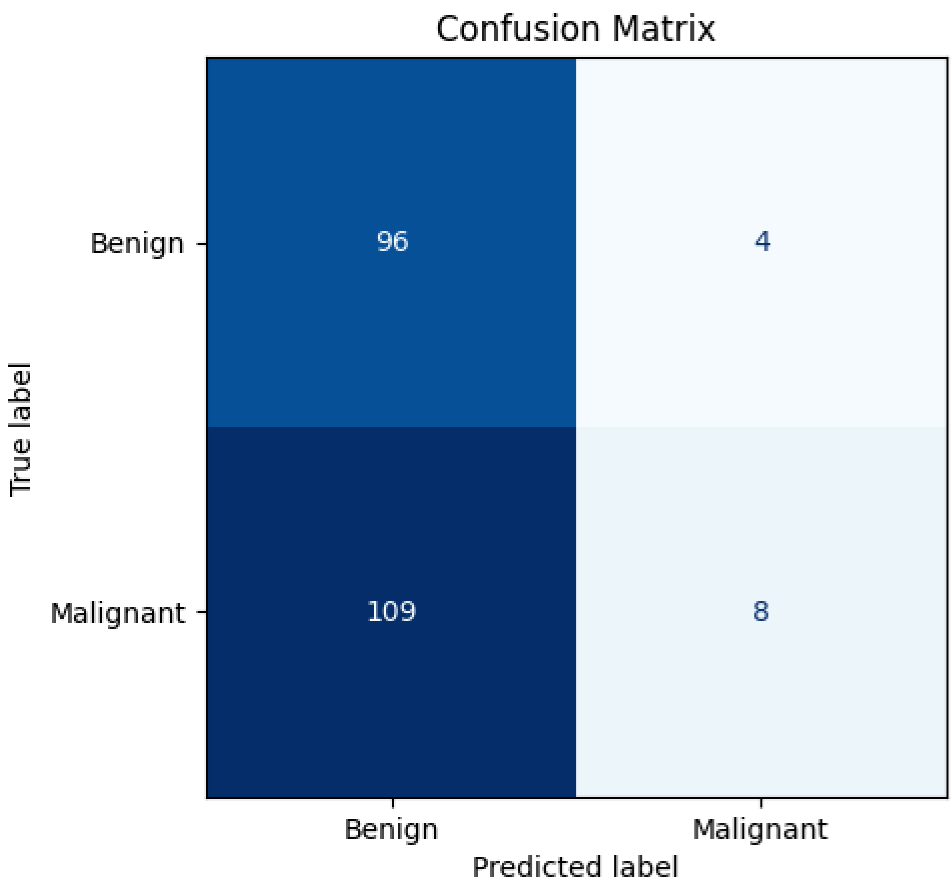
Model 3

Layer (type)	Output Shape	Param #
conv2d_5 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_5 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_6 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_6 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_7 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_7 (MaxPooling2D)	(None, 26, 26, 128)	0
conv2d_8 (Conv2D)	(None, 24, 24, 256)	295,168
max_pooling2d_8 (MaxPooling2D)	(None, 12, 12, 256)	0
global_average_pooling2d_2 (GlobalAveragePooling2D)	(None, 256)	0
dense_4 (Dense)	(None, 128)	32,896
dropout_2 (Dropout)	(None, 128)	0
dense_5 (Dense)	(None, 1)	129

45 Epochs

accuracy: 0.47926267981529236,
auc: 0.654059886932373,
loss: 0.6957703828811646,
precision: 0.6666666865348816,
recall: 0.06837607175111777


Training Loss
Loss
Epoch
Training Accuracy
Accuracy
Epoch
Training AUC
AUC
Epoch
Training Recall
Recall
Epoch



	precision	recall	f1-score	support
Benign	0.47	0.96	0.63	100
Malignant	0.67	0.07	0.12	117
accuracy			0.48	217
macro avg	0.57	0.51	0.38	217
weighted avg	0.58	0.48	0.36	217

Comparing Architectures

If you’re choosing a model right now, I’d summarize it like this:

- pick architecture 1 if you want the safer, more balanced baseline
- pick architecture 2 only if your team explicitly wants the higher-sensitivity, more false-positive tradeoff

Global Average Pooling vs Global Max Pooling

- Architecture 1: average pooling clearly looked better than max pooling
- Architecture 2: average and max pooling are pretty close, but average still has a slight edge if your priority is catching malignant cases

Results

Architecture 1 @35 epochs is the best overall model
Test Metrics:

- accuracy 0.6959, auc 0.7087, precision 0.6977, recall 0.7692
- Confusion matrix:
  [[61 39]
   [27 90]]
- Classification report:
  - Benign recall: 0.61
  - Malignant recall: 0.77

Architecture 2
Test Metrics:

- accuracy 0.6636, auc 0.7362, precision 0.6549, recall 0.7949
- Confusion matrix:
  [[51 49]
   [24 93]]
- Classification report:
  - Benign recall: 0.51
  - Malignant recall: 0.79

Main Takeaways:

- model_2 is only slightly better at catching malignant cases: 93 vs 90
- but model_1 is noticeably better on overall accuracy, precision, and benign handling
- model_1 makes fewer total mistakes: 66 vs 73
- model_1 is more balanced across both classes

Appendix

Image Sizing

224 x 224 is mostly a practical default.

Why it’s commonly used:

- it’s large enough to preserve useful visual structure
- it’s small enough to train reasonably fast
- many image-classification examples and pretrained models use 224 x 224

For your CNN, it gives a good balance between:

- image detail
- memory usage
- training speed

Why not much bigger, like 512 x 512?

- more compute
- more memory
- slower training
- higher risk of overfitting, especially with a modest dataset

Why not much smaller, like 64 x 64?

- you may lose fine lesion details
- texture, border irregularity, and color variation can get blurred out

So 224 x 224 is basically a strong starting compromise.

It was not chosen because it is magically optimal for melanoma. It was chosen because:

- it is standard
- it is manageable
- it usually works well as a first pass